

Balancing College and Work Activities: The Gig Economy Meets Online Education*

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Abstract

Could flexible learning and work environments mitigate the trade-offs that college students face while working? We answer this question using unique data from a partnership between Arizona State University and Uber that allows eligible drivers to take free online courses. Guided by a simple analytical framework, using high-frequency data and an instrumental variables strategy, we analyze how labor supply and study effort respond to changes in work and school demands. We show that a 10% increase in weekly coursework reduces Uber hours by about 1%, indicating limited short-run displacement between coursework and labor supply in this setting. Study time also remains mostly unaffected by fluctuations in labor market conditions. Survey evidence indicates that students place substantial value on flexible work and education formats and expect meaningful income gains from completing an online degree. Overall, our findings suggest that, in this flexible work-learning context, students accommodate short-run work-study trade-offs with relatively limited disruption.

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1 Introduction

Student employment is a central feature of higher education in the United States. In 2019, 45% of full-time and 84% of part-time undergraduate students worked for pay (US Department of Education, 2020). Among working students, employment was also intensive: 64% of full-time and 86% of part-time undergraduates worked at least 20 hours per week. These patterns are not new. The average labor supply of 18- to 22-year-old full-time undergraduates nearly doubled between 1970 and 2000 (Scott-Clayton, 2012), and high levels of student employment are likely to remain an important feature of college attendance. Two broader forces point in this direction. First, the cost of college has risen substantially, increasing the need for many students to finance their education through work.¹ Second, technological change has altered labor-market demands, reducing opportunities in low-skill and routine jobs and encouraging workers to acquire new skills while remaining employed (Saliola et al., 2020). The rise of artificial intelligence may reinforce this pressure. Given that AI is expected to affect nearly 40% of jobs worldwide (Cazzaniga et al., 2024), even many high-skill workers may need to update their skills in response to changing skill demands.

At the same time, existing evidence suggests that working while attending college meaningfully crowds out schooling time, which often negatively affects college performance (Stinebrickner and Stinebrickner, 2003; Beffy et al., 2010; Kalenkoski and Pabilonia, 2010; Scott-Clayton, 2011, 2012; Darolia, 2014; Scott-Clayton and Minaya, 2016; Soliz and Long, 2016; Neyt et al., 2019; Le Barbanchon et al., 2020). Taken together, these facts highlight the potential value of new alternatives that could help students balance both activities.

The aim of this study is twofold. First, we examine how students allocate time between coursework and paid work in a setting where both activities offer substantial scheduling flexibility. This setting allows us to examine whether short-run, within-semester work-study trade-offs remain economically meaningful when students can adjust both work and study effort at high frequency. We quantify these trade-offs by estimating cross-elasticities between course demands and labor supply, and between earnings opportunities and study effort, in a unique setting where online learning is combined with flexible work. A large literature

¹Between 2006 and 2020, after adjusting for inflation, the average cost of attendance at public and non-profit private four-year universities increased by almost 20% (Ma and Pender, 2022).

studies how student work affects academic performance; however, little is known about how college demands affect labor supply, or how short-run earnings opportunities affect study effort. Thus, our primary contribution is to provide the first high-frequency estimates of these trade-offs, exploiting weekly within-individual variation in coursework and local labor market-level wages.

Second, we turn from short-run time allocation to the broader value students attach to this highly flexible model of work and schooling. Using survey evidence, we examine students' expected returns to completing an online bachelor's degree, their stated likelihood of enrolling under alternative combinations of learning modality, work-schedule flexibility, and tuition cost, as well as their prior attachment to higher education. We then place these survey-based measures alongside administrative records on credit accumulation and graduation, which provide evidence on students' realized progress toward degree completion.

We study these two margins in the context of a partnership between Arizona State University (ASU) and Uber, which allows eligible drivers to enroll in ASU Online classes tuition-free; henceforth, we refer to drivers enrolled in the ASU program as Uber-students.² The partnership combines the two forms of flexibility that are central to our question: students can complete online coursework with substantial scheduling flexibility while also adjusting their Uber labor supply at high frequency.

Although this is a specific institutional setting, the underlying combination of flexible work and flexible online learning lies at the intersection of broader changes in both education and work. The share of postsecondary students taking distance courses rose from 28% in 2014 to 37% in 2019 (U.S. Department of Education, National Center for Education Statistics, 2020), while the gig economy workforce expanded by more than 300% between 2015 and 2021 (Denes et al., 2023). Together, these trends suggest that settings in which students combine flexible work and flexible online learning may become increasingly important in higher education. Thus, the ASU–Uber partnership is not only useful for measurement and identification; it also provides evidence on a work-study arrangement that may become increasingly common. This dual flexibility may be especially consequential for students

²Programs in which companies fund their employees' education have become increasingly common in recent years. For instance, Starbucks, Walmart, and Disney also cover the tuition costs for employees pursuing post-secondary education.

whose enrollment decisions are tightly linked to work constraints, including nontraditional and older students, as well as students from lower socioeconomic status (SES) backgrounds who are more likely to work while enrolled (Chen and Nunnery, 2019).

Estimating these trade-offs is empirically challenging because work hours and study effort are rarely measured together at high frequency, even though the relevant trade-offs often occur within weeks or semesters. Moreover, both activities are chosen jointly, making it difficult to separate responses to coursework demands from responses to labor market opportunities. Three key features of our institutional setting allow us to address these challenges. First, Uber-students face a flexible environment where work and study hours can be adjusted frequently. This makes the setting well-suited for studying whether short-run work-study trade-offs remain meaningful when students have more flexibility to re-time work and study effort. Second, the Uber and ASU Online platforms provide granular, high-frequency information about drivers and students over time, offering a unique opportunity to examine how effort responds to changes in work and college conditions. Third, the degree of complementarity between Uber work and skills acquired in college is likely to be much lower than in other work-study contexts. In other words, college courses are unlikely to improve a student’s capabilities as an Uber driver, making the interpretation of our findings cleaner.

For the main high-frequency analysis, we link records from the Uber and ASU Online platforms. The Uber records cover drivers enrolled in fully online BA degree-seeking programs at ASU, as well as a subset of their Uber-driving peers who qualified for but did not enroll in the ASU program. These records include weekly information on completed trips, total pay, minutes on the application, driving habits, city-level labor market conditions, and background characteristics. The ASU records cover the universe of Uber-students and their online classmates who are not program participants. They include student background characteristics, transcript data, and weekly activity measures from the online learning platform, including points earned, assignments submitted, Canvas logins, clicks, and minutes online. Together, these linked records allow us to measure weekly work and study effort for the same individuals at high frequency over time.

Our econometric specifications are derived from a simple analytical framework that justifies our empirical models and helps characterize the costs of performing learning and work-

ing activities simultaneously. To address selection and endogeneity concerns, our empirical strategy combines within-individual variation, classroom-level variation in course demands, and instrumental variables based on local labor-market conditions to identify the short-run trade-offs associated with balancing work and learning activities. More specifically, we exploit weekly-level variation in the Uber and ASU platforms and plausibly exogenous changes in labor market demand and course-week learning tasks (i.e., course demands) to identify key cross-elasticities.³

Our main high-frequency results indicate that average weekly course demand decreases weekly driving time by approximately 1.7 hours. More specifically, a 10% increase in average online learning activities decreases average driving time by approximately 1%, where driving time refers to the number of hours that drivers spend connected to the Uber platform.⁴ Turning to labor market conditions, we find that a 10% increase in market-level hourly pay decreases study time for the average Uber-student by only about 1.7%, indicating that study time is relatively unresponsive to short-run changes in labor market conditions. Overall, these cross-elasticities indicate limited short-run, within-semester displacement between coursework and Uber labor supply in this highly flexible setting. Consistent with these findings, we show that Uber-students actively reallocate their driving schedules when coursework becomes more demanding, while we find no meaningful evidence that coursework reduces work quality or that driving harms academic performance.

We then turn to students’ perceived and realized returns from the ASU–Uber program. Survey respondents expect substantial income gains from completing an ASU Online degree, report a higher likelihood of enrolling in a bachelor’s degree in scenarios that combine online

³Given that our data come from drivers in a large number of cities, changes in labor demand could be driven, for example, by different social events that may happen in these cities. We show that variations in hotel occupancy rates are consistent predictors of labor market pay at the city-week level, even after including a large set of controls.

⁴We interpret our elasticity estimates as short-run, within-semester substitution responses and use them to characterize the extent to which flexible work and coursework allow students to reallocate time within the week. We also report the implied level effects. For instance, translating our baseline estimated elasticity into levels implies that an additional hour of study is associated with about a 40-minute reduction in Uber driving time. However, because baseline study time is low relative to baseline work time (2.45 vs. 18.25 hours per week), and because weekly hours are bounded below by zero with many observations near that boundary, a fixed one-hour change does not represent a comparable relative response across the work and study margins. For this reason, we emphasize elasticities, which express responses to proportional changes around observed baselines. The level and elasticity summaries are re-scalings of the same local effect, and we present both.

learning with flexible work arrangements, and often entered the program after a period of non-enrollment in higher education. Administrative evidence on degree progress shows that among the program’s first four entry cohorts, 27.2% of entrants have already graduated, higher than the 24% six-year graduation (approx.) benchmark for ASU’s overall Online undergraduate population, and the average entrant has accumulated about 81 ASU credits, roughly two-thirds of the credits required for a bachelor’s degree. Combining these progress measures with our estimated opportunity costs, we estimate that, under a plausible 15% post-graduation wage gain, the average entrant would offset expected opportunity costs after roughly five years.

Our findings indicate that, in this highly flexible setting, short-run work-study trade-offs are modest and students actively use scheduling flexibility to adjust to changing course and labor market demands. The estimates are therefore informative about within-semester time allocation when both work and coursework can be adjusted at high frequency. While these results should not be used to make strong claims about less flexible settings, they do suggest that pairing flexible work with online learning may help some students combine schooling and employment with relatively limited short-run disruption.

2 Institutional Details: The ASU-Uber Partnership

Beginning in the fall semester of 2018, ASU and Uber initiated a partnership to make higher education more accessible to drivers who meet activity, tenure, and performance requirements on the Uber platform. Under the partnership, Uber covers tuition for drivers in the U.S. who have completed at least 3,000 rides and satisfy ongoing driving and ratings requirements.⁵ With the guarantee of tuition coverage, participants can choose among several educational options. The option most relevant for our analysis is enrollment in a fully online undergraduate degree program. Notably, ASU does not distinguish between online and in-person degrees on students’ transcripts or diplomas, so the degree option may provide a particularly valuable credential for drivers seeking to combine undergraduate education with flexible work.

⁵The program also extends to beneficiaries of the drivers, such as a spouse, child, sibling, or parent. However, our analysis focuses only on drivers. Approximately 30% of the program participants are family members of the Uber drivers.

Alternatively, participants may enroll in shorter (arguably lower-commitment) non-degree options, including entrepreneurship or English-language competency courses, though we do not focus on these programs.

Qualifying Uber participants pursuing a fully online undergraduate degree at ASU have over 140 degree programs to choose from. The courses are offered following the standard university calendar, which includes full-term courses (Session C) and half-term courses that run through the first half (Session A) or second half (Session B) of the term. However, within a course, students generally have considerable flexibility over when they watch lecture videos and, depending on the course and instructor, some additional flexibility in completing assignments and taking exams. Ultimately, the exact level of flexibility depends on the course and instructor preferences: some courses may allow students to access problem sets or exams at any time, whereas others may only allow access in given weeks or days around assignment due dates. Overall, the flexible structure of these courses is important for our analysis because online enrollment gives students substantial discretion over when to complete coursework, while the academic calendar and assignment deadlines still generate predictable variation in course demands over time.

Although the 3,000-ride threshold may appear substantial, observed driving rates in our sample suggest that many active drivers could reach this threshold within a relatively short period. For example, Uber-students in our sample complete an average of roughly 43 weekly trips when they are not actively enrolled in classes. At that rate, drivers would surpass the necessary 3,000 trips in less than a year and a half, and a more casual driver who completes only 30 trips per week would surpass the qualification in under two years. Once this goal is met, drivers must also have reached either Uber Pro Gold, Platinum, or Diamond status. To first achieve Gold status, a driver must meet a location-specific point goal, have an average star rating of 4.85 or greater, and have a cancellation rate (conditional on trip acceptance) of under 4%.⁶ Once one of these different Uber statuses has been attained, the driver must

⁶The particulars of the point-based goals depend on location, but we can provide some general insights. These points are earned through the completion of rides, with the number of points awarded per completed trip being one or more, contingent upon the time of day and other factors. In most areas (with some exceptions), drivers must accrue between 200 and 600 points within a three-month time frame to sustain their status. As an illustrative example, assuming a midpoint target of 400 points and 1.5 points per trip, our rough calculations suggest that drivers would need to complete less than 21 trips each week to meet this goal. These requirements appear unlikely to bind for most drivers in our sample: full-time drivers (i.e.,

continue to reach their location-specific point total, receive a minimum star rating of 4.75, and maintain a cancellation rate below 10% in order to keep their Uber Pro status. The Uber Pro Gold, Platinum, or Diamond status must be held in three-month periods, such that the activity in the previous three months determines the status in the current period.⁷

3 Data Description

Our analysis combines high-frequency administrative data from ASU and Uber with original survey data and aggregate longer-run educational outcomes. Data provided by ASU enables us to monitor the interactions of Uber-students and their classmates with each course’s online platform on a weekly basis. These ASU data include weekly assignment performance and overall course outcomes. We complement the student activity and performance data with Uber administrative data covering drivers who participate in the ASU-Uber partnership, excluding family-member beneficiaries. These data also include a subset of eligible drivers for the ASU-Uber program who chose not to enroll. Notably, this data set represents student-drivers nationwide, not just those residing in Arizona. Additionally, we have access to data on market-level demand in each driver’s local labor market. We merged these two data sets at the individual-week level. For our empirical analysis, we omitted the peak period of the COVID-19 outbreak, as the pandemic significantly disrupted Uber activities. Consequently, our sample spans from August 2018 to March 2020, and from January 2021 to May 2021. We choose to include the Spring 2021 term in our analysis because, by that point, both Uber driving behaviors and ASU Online activity had roughly returned to pre-pandemic levels. However, our primary findings remain qualitatively similar when the analysis is confined to the pre-COVID-19 period.

To evaluate program participants’ subjective valuation of school and work flexibility, in November 2023, we conducted a survey of (currently enrolled) Uber-students and a random subset of their classmates to augment the data from the ASU and Uber platforms. This

drivers who average at least 25 hours driving per week when not enrolled in ASU courses) complete about 49 trips each week, on average; part-time drivers complete about 28 trips per week.

⁷The following link provides further details concerning the program’s characteristics: <https://uber.asu.edu/>.

survey gathered a host of additional information of interest, including allocation of time in activities beyond the platforms, expectations regarding labor market and graduation outcomes, and preferences for flexible education and learning formats. Unfortunately, the survey data cannot be integrated with the Uber and ASU data sets. Finally, we obtained aggregate graduation and credit-accumulation statistics through December 2025 for the first four cohorts of ASU-Uber program participants, from which our analytic sample is derived. As with the survey, we cannot directly link these outcomes to individuals in our high-frequency analysis; but these data provide valuable insights into longer-term credit accumulation and graduation rates for program participants.

We next present summary statistics and key features of the ASU and Uber data, and also briefly describe our survey design. We defer discussion of the aggregate longer-run administrative records on credit accumulation and graduation to Section 9. Appendix A provides further details on sample construction and variable definitions.

3.1 ASU Data

The first data set used in our analysis comes from ASU students enrolled in fully online undergraduate programs. Because the degree programs available through the ASU-Uber partnership are operated through ASU Online, students interact with most course materials through Canvas, ASU’s online learning management system. This feature allows us to observe high-frequency measures of student engagement at the student-course-week level.

3.1.1 Characterizing ASU-Uber Students

Background Characteristics Table 1 shows background characteristics and proxies for academic preparation of Uber-students and their classmates enrolled in ASU Online undergraduate programs. The first set of columns reports means and standard deviations for Uber-students, while the second reports the same statistics for their classmates who are not ASU-Uber program participants. Throughout the remainder of the text, we denote this group of students as “class peers”. Uber-students differ in multiple dimensions from their classmates. They are, on average, 11 years older, overwhelmingly male (85%), and more

Table 1: Student level observables, by ASU-Uber program status

	Uber-Students		Class Peers		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
<i>Background Characteristics</i>					
Age	39.88	(10.044)	28.51	(8.023)	11.37***
Female	0.15	(0.361)	0.61	(0.489)	-0.45***
White	0.36	(0.480)	0.58	(0.493)	-0.22***
Black	0.33	(0.470)	0.07	(0.256)	0.26***
Hispanic	0.15	(0.353)	0.21	(0.409)	-0.07***
Asian	0.07	(0.251)	0.05	(0.211)	0.02***
First generation	0.57	(0.495)	0.39	(0.489)	0.17***
AZ resident	0.06	(0.238)	0.23	(0.422)	-0.17***
<i>Financial need</i>					
None	0.00	(0.057)	0.01	(0.097)	-0.01***
Very low	0.03	(0.172)	0.06	(0.232)	-0.03***
Low	0.08	(0.266)	0.11	(0.309)	-0.03***
Moderate	0.32	(0.468)	0.35	(0.478)	-0.03**
High	0.05	(0.225)	0.07	(0.250)	-0.01**
Very high	0.51	(0.500)	0.39	(0.487)	0.12***
Ever Pell eligible	0.50	(0.500)	0.49	(0.500)	0.01
<i>Academic Profile</i>					
Incoming GPA	2.71	(0.587)	2.88	(0.634)	-0.17***
Transfer status	0.62	(0.486)	0.72	(0.447)	-0.11***
Transfer hours	73.35	(51.569)	52.47	(39.369)	20.88***
Number of terms enrolled	3.01	(1.656)	4.64	(3.280)	-1.63***
Average credit hour load	7.95	(3.027)	8.77	(3.147)	-0.82***
STEM degree	0.40	(0.489)	0.30	(0.456)	0.10***
Observations	1540		64248		

Note: “Uber-students” refers to ASU students participating in the ASU-Uber partnership. “Class peers” corresponds to all other students enrolled in the same classes as the Uber students. First-generation refers to students whose parents did not graduate college. “AZ resident” indicates if the student is listed as an Arizona resident. Each of the financial need variables “None”, “Very low”, “Low”, “Moderate”, “High”, and “Very high” indicate the financial need status of the student, which is constructed by ASU based on students’ applications to federal student aid. “Ever Pell eligible” indicates if the student was ever eligible for a Pell grant. The variable incoming GPA corresponds to a student’s transfer GPA or, if transfer GPA is inapplicable, high school GPA. “Transfer status” is an indicator variable denoting whether the student transferred from another higher-education institution, and “transfer credit hours” refer to the number of credit hours carried over from their previous institution. “Number of terms enrolled” denotes the number of distinct terms that the student has been enrolled in ASU courses. “Average credit hour load” reflects the student’s average number of credit hours per term, and “STEM degree” indicates if the student is in a science, technology, engineering, or math degree program. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table 2: Weekly Canvas activity and outcomes, by sample

	(1)		(2)		(3)		(4)	(5)
	Uber-Students Active in Uber		Uber-Students Inactive in Uber		Class Peers		Col (1) - (3)	Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Difference	Difference
Activity								
Days connected to Canvas per course	3.57	(2.30)	3.75	(2.27)	3.48	(2.12)	0.09***	-0.18***
Clicks per course	75.97	(89.33)	79.04	(85.57)	63.94	(69.70)	12.03***	-3.07***
Hours online per course	1.57	(2.12)	1.61	(2.10)	1.25	(1.64)	0.31***	-0.04**
Outcomes								
Assignments submitted per course	1.97	(4.14)	1.99	(2.92)	2.12	(4.05)	-0.15***	-0.02
Assignment share per course (%)	90.19	(27.59)	90.55	(26.81)	91.56	(25.10)	-1.37***	-0.36
Point share per course (%)	79.40	(29.31)	80.13	(28.66)	81.16	(28.64)	-1.76***	-0.74**
Observations	43527		12201		1831331			

Note: All statistics reported in the table are at the student-by-course-by-week level. “Uber-Students: Active in Uber” denotes course-weeks in which the student shows positive driving hours and is enrolled in ASU classes. “Uber Students: Inactive in Uber” denotes weeks in which the student completes no trips but is enrolled in ASU classes. “Class peers” correspond to all other students who are enrolled in the same classes as the Uber students. Assignment and point shares refer to the total number of assignments submitted and points earned relative to the number of assignments due and points possible, respectively. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

racially diverse. For instance, 33% of the Uber-students are Black, whereas the analogous share for their peer population is only 7%. In addition, Uber-students are more likely to be first-generation (i.e., do not have a parent who graduated from college) and to reside outside Arizona. The middle panel shows that Uber-students have higher financial needs (based on FAFSA records): 51% of them are designated as “very high” financial need versus 39% of their peers. The last panel shows that Uber-students have lower incoming GPAs than their counterparts, are less likely to enter ASU as transfer students, but have accumulated more transfer credit hours on average. They have also completed fewer terms at ASU, are more likely to pursue a major in STEM, and have a lower average credit-hour load per term than their class peers. Note that most ASU Online courses that we consider in our analysis are worth 3 credit hours, so Uber-students and their class peers complete roughly 2.7 and 2.9 courses per term, respectively.

Weekly Academic Engagement Table 2 shows proxy measures for student engagement (i.e., active days, clicks, hours online) and performance in ASU Online courses at the student-course-week level. The sample is divided into three groups: Uber-students active in Uber, Uber-students inactive in Uber, and class peers, where active and inactive distinguish the weeks in which Uber-students were actively participating on the Uber platform; if they complete at least one trip during a week, they are classified as active for that week. On average, Uber-students show greater engagement on Canvas than their classmates. They

tend to spend more days per week connected to the ASU platform, record more clicks, and spend more time on it.⁸ Uber-students also slightly increase their engagement on Canvas during weeks when they are inactive on the Uber platform. Finally, Uber-students (regardless of Uber activity status) spend about 1.6 hours online per course in a given week, amounting to 2.84 hours of online study activity per week when averaging across all courses and weeks.⁹

In terms of weekly academic performance, Uber-students submit, on average, 1.97 assignments per course during active driving weeks, slightly less than their peers (2.12), representing 90.2% of the total assignments requested. They obtain approximately 80% of the total points, which is similar to the point share of their class peers (81%).¹⁰ Finally, differences in weekly academic performance between active and inactive Uber weeks are small.

To conclude, a possible concern with these data is that our various Canvas engagement variables may be incomplete measures of study time and effort. For example, time spent reading hard copies of a textbook would not be included in the measure of minutes on Canvas. However, the students’ self-reported study time allocation on and off the platforms (see subsection 3.3) and the strong correlation between Canvas hours and academic performance (discussed in Section 8) suggest that our measures of study effort constitute good proxies for actual study time. Furthermore, because our empirical specifications include individual fixed effects, they account for time-invariant differences across students in the extent to which Canvas activity captures total study effort. Remaining concerns would arise if within-

⁸Appendix Table G.1 shows that our measures of online activity (i.e., days connected to Canvas, number of logins and clicks, and time spent on the platform) are highly correlated. In particular, the high correlation between hours online and clicks suggests that idle time is not likely to be an important concern when proxying study time with “hours online”.

⁹Because direct measures of study time are rare, it is difficult to benchmark whether observed Canvas time is high or low. Using course evaluation surveys, Ahn et al. (2024) find that students at the University of Kentucky spend, on average, 2.4 hours per week studying for a non-STEM course and 3.5 hours per week studying for a STEM course. Babcock and Marks (2011) document that U.S. college students’ average weekly study time, including class time, declined from 40 hours in 1961 to 27 hours in 2003. These benchmarks suggest that our Canvas-based measure captures only part of total academic effort, which may also include reading, studying, or completing work off platform.

¹⁰The point share is determined by dividing the points earned in a course for a given week by the points possible for the student in a given course-week. Assignments submitted refer to the number of assignments the student submitted throughout the week (assignments do not necessarily need to be submitted in the week they are due). In Appendix Table G.2, we provide an alternative version of Table 2 where we re-weight the control group by coarsened exact matching (see Iacus et al., 2012) on many of the observable characteristics presented in Table 1. After doing so, the Uber-students’ Canvas activity measures are much more similar to the control group, suggesting that once we control for fixed observable characteristics, the Uber-students’ study behaviors are quite similar to their non-Uber peers.

Table 3: Course-level summary statistics, by sample

	(1)		(2)		(3)
	Uber Students		Class Peers		Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Difference
<i>Activity</i>					
Total hours on Canvas	14.63	(12.73)	11.47	(9.62)	3.16***
Assignments submitted	18.38	(16.89)	19.37	(17.54)	-1.00***
<i>Outcomes</i>					
Passed course	0.76	(0.43)	0.83	(0.38)	-0.07***
Numeric course grade	2.49	(1.66)	2.70	(1.54)	-0.21***
Observations	6591		213842		

Note: “Uber-Students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students that are enrolled in the same classes as the Uber students. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

student changes in off-platform study time are systematically correlated with weekly Uber activity.

Course-Level Academic Outcomes Table 3 reports course-level measures of Canvas activity and academic outcomes for Uber students and their class peers. The top panel summarizes total activity over the course, while the bottom panel reports course completion and grade outcomes. On average, Uber students spend approximately 14.6 hours on Canvas per course, compared with 11.5 hours among their class peers. For reference, most courses last seven to eight weeks, though a small number of courses run for 15–16 weeks. Uber-students submit roughly 18.4 assignments per course, whereas their classmates submit approximately 19.4. These patterns suggest that Uber students remain actively engaged with course materials, despite the additional work demands they may face outside of school.

The bottom panel of Table 3 shows that Uber-students have a 76% passing rate and average course grades of 2.49. Unlike activity, these measures of academic performance are somewhat worse compared to their peers (i.e., a passing rate of 83% and an average grade of 2.70).¹¹ However, these differences are somewhat expected given that enrolled Uber-

¹¹Appendix Table G.3 also presents summary statistics regarding the progress of Uber-students and their class peers (i.e., cumulative GPA, and credits to date). Additionally, Appendix Table G.4 presents the same summary statistics as in Table 3, but when we re-weight the “Class Peers” to be more similar to the Uber-students on observables. Upon re-weighting, the differences in activity across groups attenuate considerably.

students, on average, are from less advantaged backgrounds and have substantially lower prior academic credentials. In light of the evidence presented in this table, the higher levels of engagement presented in Table 2 may reflect greater effort by Uber students to keep pace with course material.

Overall, the main takeaway from these summary statistics is that Uber-students, despite coming from more disadvantaged backgrounds than their class peers, have only slightly lower course completion rates and grades but somewhat higher levels of engagement in learning activities than their classmates.

3.1.2 Within-Term Variation in Canvas Engagement

To further characterize within-term variation in course effort, we estimate a descriptive event-time specification for Uber students’ weekly Canvas hours; this exercise is intended to describe the time profile of Canvas engagement over the academic term, and should not be interpreted as causal. Specifically, we analyze how study time in the different weeks of the term compares to the third week before the beginning of the academic semester.¹²

The results for eight-week courses, the modal course length in our sample, are presented in Figure 1. As expected, Canvas hours are essentially zero in the weeks leading up to the beginning of the course.¹³ Canvas hours increase sharply to roughly three hours per week in

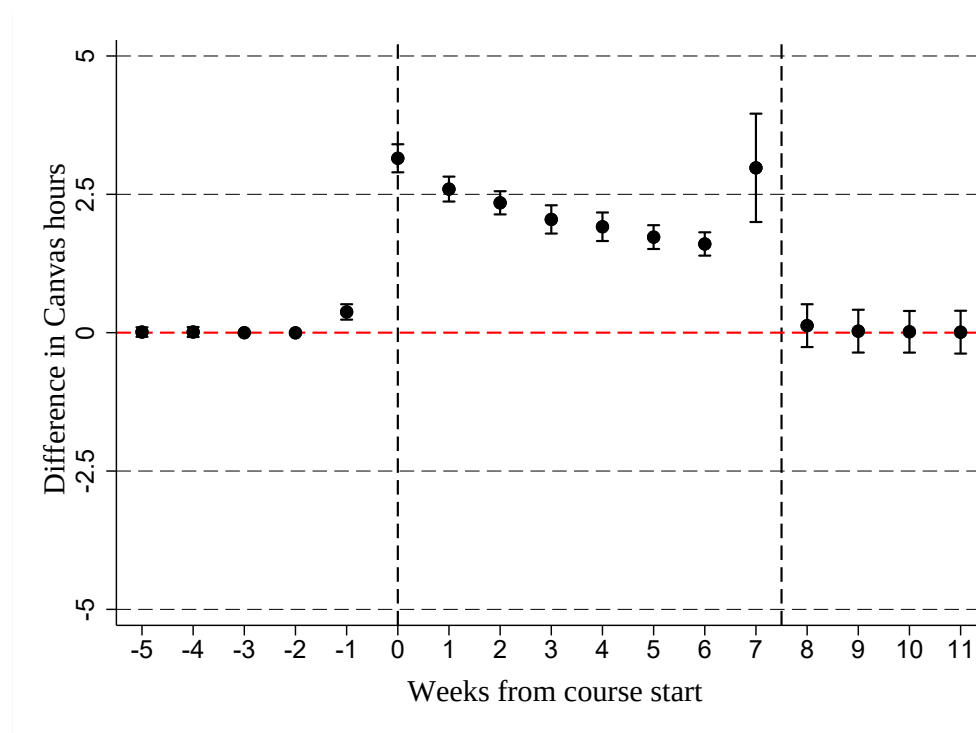
¹²We estimate a specification of the following form:

$$\begin{aligned} CourseE_{i,c,t} = \sum_{\substack{y=-5 \\ y \neq -3}}^T \alpha_y \mathbb{I}(t - t_c^* = y) + \beta_1 CourseWork_{i,t,-c} + \beta_2 MarketHourlyPay_{l,t} + \beta_3 Weather_{l,t} \\ + X_{i,t}\Phi + \psi_i + \delta_m + \phi_c + \varepsilon_{i,c,t}, \end{aligned} \quad (1)$$

where $CourseE_{i,c,t}$ is Canvas hours for Uber-student i , taking course c , in week t . $CourseWork_{i,t,-c}$ captures course peer activities in the other courses ($-c$) in which the student is enrolled (controlling or not for this variable provides very similar results). The vector $X_{i,t}$ includes controls for the number of course days in the week and an indicator for whether the week includes Christmas. ψ_i , δ_m , and ϕ_c denote individual, month-year and classroom fixed effects, respectively. $MarketHourlyPay_{l,t}$ and $Weather_{l,t}$ denote the average market-level hourly pay and second-order polynomials of rainfall and snowfall, respectively, in the labor market l for individual i in week t . $\varepsilon_{i,c,t}$ corresponds to the idiosyncratic shock. Finally, $\mathbb{I}(t - t_c^* = y)$ denotes indicators for the week of the course, y , where t_c^* is the first week of the course ($y = 0$). The excluded week is the week that is three weeks prior to the start of the course. Thus, each $\hat{\alpha}_y$ captures average Canvas hours in event week y , relative to the third week before the course starts, conditional on the included controls and fixed effects. Section 5 discusses the rationale for this specification. To appropriately capture the evolution of driving hours and study hours, we exclude individuals who drop out or withdraw at some point during the course.

¹³The point estimate for week -1 is slightly above zero because some courses allow students to access the

Figure 1: Variation in study time during eight-week courses



Notes:—OLS coefficient estimates from a version of Equation (1) where the dependent variable is hours on Canvas and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

the first week of class, decline gradually over the middle of the course, and then rise again in the final week. This pattern suggests that students devote substantial time to Canvas at the start of the course, likely as they orient themselves to course requirements and initial assignments, before reducing weekly engagement during the middle of the term. The final-week increase likely reflects end-of-course assignments, exams, or other culminating course requirements.

3.2 Uber Data

The Uber driver data include background characteristics and individual weekly driving activity corresponding to two groups of drivers: Uber-students and a random sample of drivers who qualify to participate in the ASU-Uber partnership program but did not enroll (henceforth, Uber peers). Additionally, we have access to city-level labor market characteristics and weather conditions.¹⁴

3.2.1 Characterizing ASU-Uber Drivers

Background Characteristics Table 4 shows how Uber-students compare with Uber peers; Uber peers are drivers who qualify for the ASU-Uber program but do not enroll in ASU courses. Uber-students are, on average, more likely to be female and younger than their Uber peers. Uber-students are also more likely to report English as their primary language, less likely to report Spanish as their primary language, and more likely to work in Arizona than their driving counterparts (i.e., 6% vs. 2%). The bottom panel of this table also indicates that Uber-students have longer Uber tenures than their peers. In particular, they have been driving for more than 128 weeks, and during 79% of those weeks (i.e., 102 weeks), they have been active on the Uber platform. On average, they have completed more than 8,100 trips, indicating an important degree of attachment to driving with Uber. These activity levels suggest substantial attachment to the Uber platform and indicate that driving is likely an economically meaningful activity for many participants. Finally, Uber-

Canvas page before the official start of the term. This is why we use week -3 as the omitted category.

¹⁴The drivers in our sample work in 111 distinct geographic regions/cities across the United States. Information on weather conditions was downloaded from the PRISM Climate Group at Oregon State University.

Table 4: Driver-level observables, by sample

	(1)		(2)		(3)
	Uber-Students		Uber Peers		Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Difference
<i>Background Characteristics</i>					
Female	0.15	0.353	0.10	0.300	0.05***
Age [†]	43.74	10.68	50.63	12.83	-6.89***
English	0.99	0.072	0.91	0.292	0.09***
Spanish	0.00	0.051	0.07	0.254	-0.07***
Ever driven in AZ	0.06	0.237	0.02	0.149	0.04***
<i>Uber activity</i>					
Weeks in Uber data	128.91	27.568	111.75	37.757	17.16***
Active weeks in Uber data (%)	0.79	0.159	0.81	0.170	-0.02***
Lifetime completed trips	8118.40	4554.433	7798.29	4732.587	320.11***
Lifetime average rating	4.92	0.039	4.90	0.050	0.02***
Observations	1540		96513		

Note: “Uber students” denotes ASU students who participate in the ASU-Uber partnership. “Uber peers” denotes drivers who qualify for the ASU-Uber partnership but do not enroll in ASU courses. [†]Age is based on decade of birth rather than exact year of birth, as this is how Uber released the information. English and Spanish refer to the driver’s primary language. “Active weeks” refer to weeks in which driving hours are non-zero, and “lifetime” refers to the driver’s full history on the Uber platform. “Lifetime average rating” ranges from zero to five. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

students have, on average, high Uber ratings consistent with the eligibility requirements of the ASU-Uber program.

Weekly Driving Engagement Table 5 reports weekly summary statistics on Uber driving activity during weeks when Uber students are enrolled in ASU classes. The top panel shows that Uber students spend, on average, 18.25 hours per week on the Uber platform and complete 36.62 trips.¹⁵ The bottom panel shows that Uber students earn roughly \$442 per week, corresponding to average earnings of \$12.37 per trip and \$24.26 per hour. The final rows of Table 5 show that tips account for approximately 10% of Uber students’ earnings per trip and per hour, and that these drivers maintain very high average ratings.

Appendix Table G.5 reports correlations between Uber activity measures—including completed trips and time on the platform—and measures of total pay, incentive pay, and tips. Hours on the platform and completed trips are highly correlated, suggesting that total platform time is a useful proxy for driving engagement, even though it includes some inactive

¹⁵This measure of Uber hours includes both active time (i.e., driving to pick up or complete a ride) and inactive time (i.e., waiting for a pickup request). In the individual-level data, we observe only total time on the platform.

Table 5: Weekly Uber-student driving activity

	Mean	Std. Dev.
<i>Activity and trips</i>		
Hours online	18.25	(17.29)
Completed trips	36.62	(35.74)
<i>Uber earnings</i>		
Weekly earnings	442.21	(461.71)
Earnings per trip	12.37	(5.67)
Earnings per hour	24.26	(9.89)
Tips per trip	1.12	(1.10)
Tips per hour	2.20	(2.25)
Average weekly rating	4.94	(0.13)
Observations	28896	

Note: Sample is limited to weeks in which drivers are enrolled in ASU classes. The variables “hours online”, “completed trips”, and “weekly earnings” include zeros for weeks that the drivers were inactive on the Uber platform. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

waiting time.

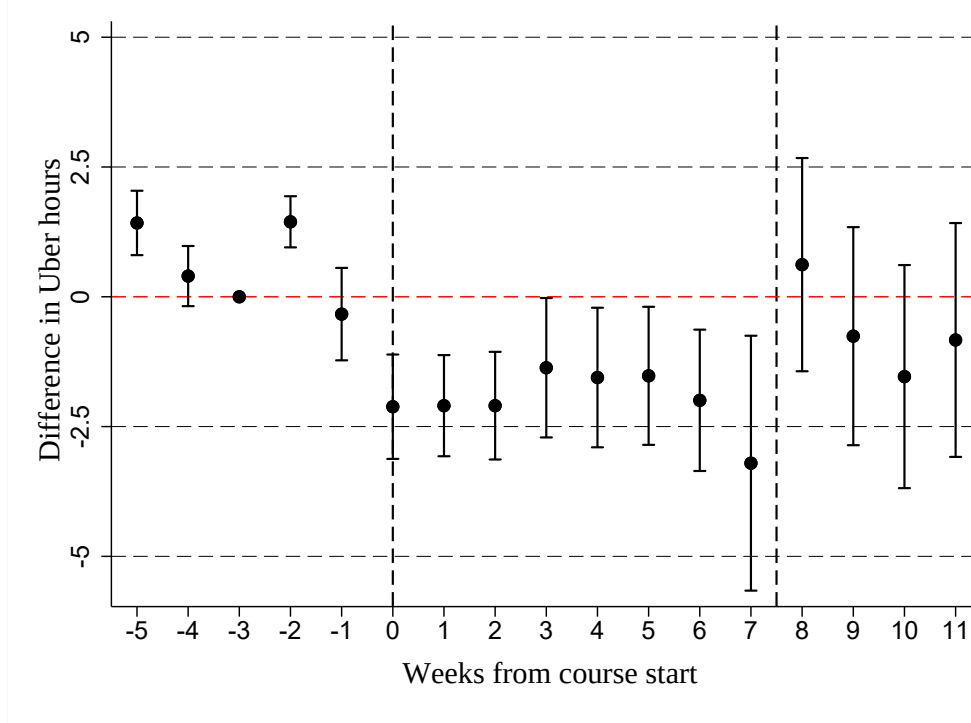
3.2.2 Variation in Weekly Uber Driving

As with study time, we complement the summary statistics in Table 5 by examining how driving hours evolve over the course of the term. We estimate a descriptive event-time specification in which the omitted category is the third week before the course begins.¹⁶ Figure 2 plots the estimated event-week coefficients for driving hours in eight-week courses.

Driving hours decline slightly in the week immediately before the course begins and fall by roughly two hours per week once the course starts. This decline remains fairly stable over the term, with a somewhat larger reduction in the final week of the course. The final-week decline likely reflects final exams, end-of-course assignments, or other culminating course requirements. After the course ends, driving hours are no longer statistically distinguishable from pre-course levels. Although Figures 1 and 2 suggest offsetting movements in Canvas engagement and Uber driving, we do not interpret these descriptive patterns as evidence of hour-for-hour substitution between work and study time. Rather, they are informative

¹⁶The empirical specification follows Equation (6) from Section 5, replacing the coursework covariate with indicators for event week relative to the start of the course, analogous to Equation (1).

Figure 2: Variation in driving hours during eight-week courses



Notes.—OLS coefficient estimates from a version of Equation (1) where the dependent variable is hours on the Uber platform and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

about short-run, within-semester reallocations that occur while Uber-students are enrolled in ASU Online courses, and should not be interpreted as cumulative year-level crowd-out of work or schooling. Additionally, given the substantial difference in baseline work and study hours, in Section 6 we present our main estimates as short-run Frisch-style elasticities, which place the two margins on a common scale.

3.3 Survey Data

All participants of the ASU-Uber program enrolled in Arizona State University’s online courses in November 2023 received an email invitation to participate in our survey. A randomly selected, equal number of their course peers also received an email invitation to participate. The survey aimed to collect data on students’ demographic backgrounds, previous academic experiences, expectations for college and labor market outcomes, time

allocation outside the online platforms, and their willingness to pay for flexible learning and working options. As an incentive for participation, respondents entered into a lottery, offering a chance to win one of 250 \$15 Amazon gift cards or one of 10 \$100 Amazon gift cards. Out of 5,478 individuals surveyed, 692 (13%) responded. The respondents included 243 Uber-students, 105 family members of Uber drivers participating in the program, and 344 classroom peers. Because our analysis does not focus on family members, their responses are excluded. The median time taken to complete the survey was 22 minutes.

Appendix Table G.6 presents a comparison of the demographics and field of study between Uber-students (excluding family members) in our survey sample and those recorded in the administrative data (i.e., platform data). This comparative analysis reveals that the demographic characteristics of the two groups are broadly similar.¹⁷ The primary deviation is observed in the female participation rate. However, this discrepancy arises because the survey also encompasses Uber Eats drivers, representing a more gender-balanced group.

The survey also sought to determine the proportion of study time Uber-students spend on the platforms and whether they drive for other ride-sharing services. Survey results reveal that 77% of the Uber-students' study time is conducted on the Canvas platform, and they dedicate over 83% of their ridesharing hours exclusively to Uber. These shares are consistent with Hyman et al. (2020), which shows that less than one-third of drivers use both the Uber and Lyft platforms in Seattle. These responses suggest that the ASU and Uber platform data capture a substantial share of students' academic engagement and rideshare work activity. Nevertheless, the platform data do not capture all off-platform study time or all sources of income, so our measures may understate students' total academic effort and economic activity.

4 Analytical Framework

In this section, we present a simple framework in which students allocate effort between work and coursework in an environment where both labor-market incentives and academic

¹⁷Appendix Figure H.1 shows that approximately half of Uber-students were either not enrolled or on a break from another institution prior to joining ASU. In addition, approximately half of the Uber-students who were previously enrolled or taking a leave from college were attending 2-year institutions.

demands vary across weeks. The model is intentionally stylized, but it serves two purposes. First, it clarifies why our empirical specifications focus on exogenous shifts in labor-market opportunities and coursework demands rather than treating work and study effort as direct causal inputs into one another. Second, it provides a structure for interpreting the estimated trade-offs between work and study time and for decomposing how jointly performing both activities affects the marginal cost of effort in Appendix D.

Agents’ Maximization Problem Equation (2) poses the problem of the agent. Individual i has linear preferences at time t (i.e., week) over earnings and academic performance, where α denotes the weight assigned to each component. Earnings are determined by the wage rate at time t and work effort (i.e., driving hours or trips completed), $w_t e_{ilt}$, while academic performance is represented by coursework grades, G_{it} (e.g., weekly college homework assignments, mid-term exams, etc.), at time t . In Equation (3), we define G_{it} as a function of individual-specific college effort, e_{ict} (i.e., study time); ability (Ab_i); and the level of difficulty of college activities (ς_t) in a given period. More formally, $G_{it} = f_i(e_{ict}, Ab_i, \varsigma_t)$. This linear specification of grades allows for (positive) complementarities in the production of grades between ability and college effort, and between the intensity of coursework requirements and college effort.

For an agent, exerting effort is costly and varies by type of activity, where λ_l and λ_c determine the cost associated with each type of effort. Finally, the model allows for (negative) complementarities in the cost of effort. For example, an additional hour of work could be more costly (λ_{lc}) in weeks when the agent exerts more effort in college activities (e.g., driving an extra hour may be more daunting in weeks when college assignments have been more time-consuming). Finally, the additional constraints reflect basic/intuitive restrictions on α and the λ s. Given these preferences, the problem of the agent is to choose the optimal vectors of effort in each activity ($\mathbf{e}_{ic}^*, \mathbf{e}_{il}^*$).¹⁸

¹⁸As mentioned earlier, the goal of the model is to provide a framework to guide the empirical specification. Thus, the model is simplified along several dimensions. For example, the decision of each period is treated as static in nature. Similarly, Uber earnings are taken as given, and we do not allow for the possibility that individuals may strategically decide to drive at times within the week with higher rates. The last simplification is partly also made because we only observe Uber earnings at the weekly level.

$$\begin{aligned}
& \max_{\mathbf{e}_{il}^*, \mathbf{e}_{ic}^*} \sum_{t=1}^T \left(\alpha w_t e_{ilt} + (1 - \alpha) G_{it} \right) \\
& - \lambda_l \sum_{t=1}^T e_{ilt}^2 - \lambda_c \sum_{t=1}^T e_{ict}^2 - \underbrace{\lambda_{lc} \sum_{t=1}^T e_{ilt} e_{ict}}_{\text{Neg. complementarities of effort btwn. work and college activities}}, \tag{2}
\end{aligned}$$

where

$$G_{it} = \beta_0 + \beta_1 e_{ict} + \beta_2 A b_i + \beta_3 e_{ict} A b_i + \beta_4 \varsigma_t + \beta_5 e_{ict} \varsigma_t + \varepsilon_{it} \tag{3}$$

$$\alpha \in [0, 1]$$

$$\lambda_l \geq 0, \lambda_c \geq 0, \lambda_{lc} \geq 0.$$

Optimal Effort To characterize the model's comparative statics, we focus on the interior solution in which both work and study effort are positive. This abstracts from the lower-bound constraint that effort cannot be negative, but it allows us to transparently derive how optimal effort responds to changes in wages and coursework demands. The solution to the agent's problem is straightforward when we specify a linear production function for grades as in Equation (3). The first-order-conditions of the agent's problem imply that the optimal levels of work and study effort satisfy:

$$e_{ilt}^* = \frac{(\alpha - 1)\lambda_{lc}[\beta_1 + \beta_3 A b_i + \beta_5 \varsigma_t] + 2\alpha\lambda_c w_t}{4\lambda_c\lambda_l - \lambda_{lc}^2} \tag{4}$$

$$e_{ict}^* = \frac{2(1 - \alpha)\lambda_l[\beta_1 + \beta_3 A b_i + \beta_5 \varsigma_t] - \alpha\lambda_{lc} w_t}{4\lambda_c\lambda_l - \lambda_{lc}^2}. \tag{5}$$

Equations (4) and (5), while simple, are appealing and informative in several important ways. First, given that w_t and ς_t change at a high frequency in our context, incentives to exert effort on learning and working activities are likely to vary periodically. For example, as expected, the model predicts that a wage increase, $\uparrow w_t$, leads to an increase (decrease) in work (study) effort, while more demanding weeks of class, $\uparrow \varsigma_t$, lead to a likely decrease

(increase) in work (study) effort.¹⁹ The fact that, in most settings, it is not possible to directly observe effort in different activities makes our empirical context appealing to understand optimal effort decisions. Equations (4) and (5) also clarify that the solution to an agent’s maximization problem only depends on exogenous variables that are independent of the agent’s choices. This highlights why our empirical specifications do not treat realized study effort and realized work effort as direct causal inputs into one another. In the model, both are jointly chosen outcomes that respond to underlying incentives, such as wages and coursework demands.²⁰

5 Empirical Strategy

Our empirical strategy exploits week-to-week variation in course demand and labor market conditions. Thus, the coefficients in our main specifications should be interpreted as within-term, intensive-margin substitution responses. In this sense, they are Frisch-labor supply responses that characterize how students reallocate time when both work and study schedules are adjustable at a high frequency. These estimates are informative about short-run displacement and strategic scheduling, but they do not identify the cumulative annual effect of enrollment relative to non-enrollment, nor the overall long-run crowd-out of work by schooling or schooling by work.

5.1 The Impact of Learning Activities on Labor Market Supply

We use weekly individual-level data on Uber driving and ASU course engagement to estimate how coursework affects labor supply. Our baseline specification, motivated by Equation (4),

¹⁹It is expected that β_3 and β_5 are positive due to likely positive complementarities between effort and ability/college activity intensity.

²⁰Wooldridge (2015) provides a discussion on this point. In particular, Wooldridge (2015) argues that an improper use of simultaneous equations models would be to model weekly hours spent studying and weekly hours working. Each student chooses these variables simultaneously as a function of the wage that can be earned working, ability as a student, etc. Therefore, it is incorrect to specify two equations where each is a function of the other.

is:

$$\begin{aligned}
WorkE_{i,t} = & \beta_0 + \beta_1 \sum_{c \in C_{it}} CourseWork_{c,t} + \beta_2 MarketHourlyPay_{l(i),t} \\
& + \beta_3 Weather_{l(i),t} + \psi_i + \delta_m + \phi_b + \epsilon_{i,t},
\end{aligned} \tag{6}$$

where $WorkE_{i,t}$ denotes Uber labor supply for individual i in week t , measured either as total hours connected to the Uber platform or completed trips. $CourseWork_{c,t}$ captures weekly course demand in course c , measured by the average Canvas hours of non-Uber course peers.²¹ We sum this measure across all courses C_{it} in which student i is enrolled in week t . $MarketHourlyPay_{l(i),t}$ denotes average active hourly pay in week t and labor market $l(i)$ among drivers who are eligible for the ASU-Uber program but do not enroll, and $Weather_{l(i),t}$ controls for local weather conditions that may affect both Uber demand and the disutility of driving.²² The specification includes individual fixed effects ψ_i , month-year fixed effects δ_m , and course-bundle fixed effects ϕ_b , where course bundles are defined by the set of courses in which the student is enrolled during the term.

The coefficient of interest is β_1 , which captures how weekly coursework demand affects Uber labor supply among enrolled student-drivers. The identifying assumption is that, conditional on individual fixed effects, course-bundle fixed effects, month-year fixed effects, market-level pay, and weather, week-to-week variation in non-Uber peers' Canvas activity represents exogenous variation in coursework demand faced by Uber students. Because identification comes from within-student variation during enrolled terms, the estimates are less likely to be confounded by time-invariant differences across students or by pre-enrollment differences in attachment to school or work.²³ We also estimate specifications with city-week fixed effects, which absorb local labor-market shocks more flexibly but preclude separate identification of β_2 .

²¹Because ASU Online courses are asynchronous and students generally do not know one another or live in the same cities, peer collaboration in coursework is likely limited.

²²Market-level active hourly pay is calculated as average total earnings divided by average active driving hours among eligible non-enrolled drivers in a city-week. Results are robust to using a measure that divides earnings by total platform time, including inactive waiting time. See Appendix A.2 for further information on variable construction.

²³Though not a threat to identification per se, including student-drivers who have dropped out of their courses in our sample could complicate our interpretations of $\hat{\beta}_1$ and $\hat{\beta}_2$. Appendix A.1 describes how we identify likely course dropouts and shows that excluding them has little effect on the main estimates.

The coefficient β_2 captures the responsiveness of Uber labor supply to transitory changes in local hourly pay. Identifying this parameter requires that unobserved city-week shocks to Uber pay are not otherwise related to individual labor supply after conditioning on the included controls and fixed effects. Weather is a natural concern because it may affect both ride demand and the cost of driving, so all baseline specifications include flexible controls for precipitation and temperature. We also provide supporting evidence in Appendix B that market-level hourly pay responds to local demand shocks, using weekly hotel occupancy data. In particular, a one-standard-deviation increase in local hotel occupancy is associated with a 0.25 standard deviation increase in market-level hourly pay.

5.2 The Impact of Labor Market Conditions on College Effort

We next estimate how transitory labor market conditions affect course engagement. As in the labor-supply specification, our estimating equation is motivated by the optimal study-effort condition in Equation (5). The primary specification is:

$$\begin{aligned} CourseE_{i,c,t} = & \delta_0 + \delta_1 CourseWork_{i,t,c} + \delta_2 \sum_{\substack{c' \in C_{it} \\ c' \neq c}} CourseWork_{i,t,c'} \\ & + \delta_3 MarketHourlyPay_{l(i),t} + \delta_4 Weather_{l(i),t} + \delta_5 X_{c,t} + \pi_i + \alpha_m + \rho_c + \eta_{i,c,t}, \end{aligned} \quad (7)$$

where $CourseE_{i,c,t}$ denotes course-level effort of student i on class c during week t . While Equation (6) is estimated at the individual-week level, this specification is estimated at the individual-course-week-level. We measure course effort using weekly Canvas hours, clicks, or logins in course c . $CourseWork_{i,t,c}$ captures course demand in course c , measured using the average Canvas activity of class peers who are not ASU-Uber program participants. The summation over $CourseWork_{i,t,c'}$ with $c' \neq c$ captures coursework demand in the student's other enrolled courses during the same week. $MarketHourlyPay_{l(i),t}$ and $Weather_{l(i),t}$ are defined as in Equation (6). $X_{c,t}$ denotes number of course days in the week and π_i , α_m , and ρ_c denote individual, month-year and classroom fixed effects, respectively.²⁴ A classroom

²⁴We include the number of course days in the week as a control because some weeks of class (such as the first and final weeks of the course) do not start on Monday or end on Friday. As a result, Canvas activity is, in a sense, mechanically lower due to fewer days of potential activity.

is defined as a given course taught by a given instructor in a given term. Finally, $\eta_{i,c,t}$ corresponds to the idiosyncratic shock.

The parameter of interest from this empirical model is δ_3 , which indicates how sensitive college effort is to labor market conditions. The identifying assumption is that, conditional on the fixed effects, observable controls, and weather conditions, market-level hourly pay is unrelated to unobserved shocks to an individual student’s course engagement. As a robustness check, Section 6.2 presents a placebo test using ASU Online students who are not enrolled in the ASU-Uber program to test for the validity of our identifying assumption.

5.3 The Effects of Work and Study Effort on Academic Performance

Finally, we examine whether variation in work and study effort translates into differences in academic performance. We consider both weekly course performance and overall course grades. First, we estimate how weekly Uber hours affect weekly course performance using the following specification:

$$PointShare_{i,c,t} = \pi_0 + \pi_1 WorkHours_{i,t} + \pi_2 Weather_{l(i),t} + \pi_3 X_{i,c,t} + \psi_i + \delta_m + \rho_c + \varepsilon_{i,c,t}, \quad (8)$$

where $PointShare_{i,c,t}$ refers to the share of possible points, on a 0–100 scale, that student i earned in course c during week t . $WorkHours_{i,t}$ are the total number of hours that a driver spent on the Uber platform in a week, including both active and inactive time. $Weather_{l(i),t}$ denotes the standard set of weather controls, and $X_{i,c,t}$ includes controls for the number of course days in the week and an indicator for whether there were no possible points to earn in that course-week.²⁵ The remaining controls, ψ_i , δ_m , and ρ_c , denote individual, month-year, and classroom fixed-effects, respectively. We also estimate analogous specifications in which Canvas hours replace Uber hours as the measure of weekly effort.

In both cases, we focus on estimates from instrumental variables (IV) specifications that attempt to account for potential simultaneity between work or study effort and weekly course

²⁵We code weeks with no possible points as receiving the maximum share, but the results are robust to excluding those weeks, or coding them as the minimum share.

outcomes. We instrument an individual’s weekly driving hours with the average market-level hourly pay in that week for drivers who were eligible for the ASU-Uber program but did not enroll. The intuition of the instrument is that, after conditioning on observable controls and fixed effects, market-level earnings should induce variation in a given student’s driving time but are otherwise exogenous to course outcomes. Similarly, we instrument an individual student’s total Canvas time in a given week with the average total Canvas hours of their non-Uber peers in that week. The key identifying assumption in this case is that, conditional on the fixed effects and observable characteristics, peer Canvas time captures exogenous week-to-week variation in course workload on the platform, and is unrelated to other idiosyncratic shocks that might affect an individual Uber student’s performance in a given week.

Second, we examine how total Uber driving during a course relates to final course grades:

$$CourseG_{i,c} = \gamma_0 + \gamma_1 \sum_{t \in \mathcal{T}_c} WorkHours_{i,t} + \phi_c + \delta_{i,term} + \eta_{i,c}, \quad (9)$$

where $CourseG_{i,c}$ denotes student i ’s final grade in course c . $\mathcal{T}_c$ denotes the weeks spanned by course c , so the summation of $WorkHours_{i,t}$ denotes the total hours driven for Uber throughout the entire course. The terms ϕ_c and $\delta_{i,term}$ denote course (e.g., ECN 101) and individual-by-term fixed effects, respectively. Due to the inclusion of individual-by-term fixed effects, variation in performance across courses within the same academic term identifies the effect of work effort on course grades. For the specification in Equation (9), this variation arises from non-concurrent courses within a term, since concurrent courses in the same session span the same weeks and therefore inherit the same realized driving hours.

A key threat to identification in the model presented in equation (9) arises if work hours are endogenous to a student’s particular ability in a course (i.e., courses in which a student is particularly capable may have high driving hours and high achievement). In order to address this concern, we adopt the same IV strategy described above: we instrument an individual’s total driving hours throughout the course with the average market-level hourly pay for drivers who were eligible for the ASU-Uber program but did not enroll, averaged across all course weeks. This instrument isolates variation in course-level driving time induced by weekly changes in labor market conditions, while remaining plausibly exogenous to overall course

outcomes.

Third, we estimate an analogous version of Equation (9) that replaces work effort with total Canvas hours during the course.²⁶ By doing so, we can combine the estimates on college effort from this specification with those obtained from Equation (7) to provide a back-of-the-envelope calculation of the effect of labor market hourly pay on academic performance through its impact on course effort. As with work effort, study time is likely endogenous. Again, we account for this by instrumenting an individual student’s total Canvas time throughout the course with the average total Canvas hours of their non-Uber peers. Note that, unlike work hours, course-specific Canvas hours can vary across concurrent courses, so identification in the study-effort specification comes from within-term differences across both concurrent and non-concurrent courses.

6 Results

6.1 Coursework and Labor Market Supply

Table 6 presents estimates of Equation (6), which examines how weekly coursework demand affects Uber labor supply. Columns (1) and (4) indicate that a 10% increase in peer coursework is associated with roughly a 1% decline in both weekly Uber hours and completed trips. In levels, the estimate in Column (1) implies that the average weekly course load reduces Uber driving time by approximately 1.7 hours per week.²⁷ In the remainder of this section, we emphasize elasticities because baseline work and study hours differ substantially, making hour-for-hour comparisons difficult to interpret. Elasticities place the work and study margins on a common scale, though we also report the implied level changes for transparency.

Coupled with the fact that Uber-students earn roughly \$24 per hour, we interpret these findings as suggesting that the average short-run opportunity cost of college enrollment

²⁶College effort is a mediating variable for the effect of work effort on academic performance. Therefore, the effect of work effort on academic performance, once conditioning on study effort, is expected to be close to zero.

²⁷This value is obtained as $-0.680 \times 2.45 = -1.7$, where 2.45 is the average sum of peer Canvas hours across the courses in which an Uber-student is enrolled. At the course level, the analogous calculation is $-0.680 \times 1.27 = -0.86$ hours per course. Projecting peers’ Canvas hours onto Uber-students’ actual Canvas hours yields a coefficient close to one.

Table 6: Determinants of work effort

	(1) Work Hours	(2) Work Hours	(3) Work Hours	(4) Completed Trips	(5) Completed Trips	(6) Completed Trips
Sum of avg. study hours across all c, peers	-0.680*** (0.071)	-0.678*** (0.072)	-0.667*** (0.100)	-1.283*** (0.146)	-1.279*** (0.146)	-1.311*** (0.205)
Market level active hourly pay, cur- rently elig.	0.122** (0.051)	0.132** (0.051)	-	0.351** (0.112)	0.381** (0.113)	-
Obs.	27475	27404	26265	27475	27404	26265
Individual Fixed Effects	✓	✓	✓	✓	✓	✓
Course Bundle Fixed Effects	✓	✓	✓	✓	✓	✓
Month-Year Fixed effects	✓	✓		✓	✓	
Weather controls		✓			✓	
City-Week FE			✓			✓
Mean dep. var.	18.24	18.21	18.39	36.52	36.48	36.78
Market level active hourly pay, peers	33.62	33.61		33.6	33.6	
Sum avg. study hours all c, peers	2.45	2.45	2.46	2.45	2.45	2.46
Mean hourly earnings	24.23	24.22	24.38	24.23	24.22	24.38
Mean pay-per-trip	12.37	12.37	12.46	12.37	12.37	12.46
Elasticity w.r.t. peer coursework	-0.091	-0.091	-0.089	-0.086	-0.086	-0.088
Elasticity w.r.t. peer earnings	0.225	0.244		0.323	0.351	

Note: The dependent variables for Columns (1) to (3) and Columns (4) to (6) are weekly work hours and weekly completed trips, respectively. The variable “Sum of avg. study hours across all c, peers” denotes the sum of the average hours per week that peers spent on Canvas across all courses. The variable “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Course bundle fixed effects denote the set of courses taken in a given term. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Elasticities are calculated by multiplying the relevant coefficient estimates by the ratio of mean peer coursework or market-level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

is approximately \$41 per week, or \$180 per month, during enrolled weeks (or \$20.6 per week per course enrolled). Columns (2)-(3) and (5)-(6) show that our estimates are robust to controlling for weather conditions and replacing month-year fixed effects with city-week fixed effects.²⁸ Appendix Table G.7 shows that further controlling for leads and lags of the independent variables does not meaningfully change the estimated effect of study hours on work effort.

²⁸The number of observations varies slightly across columns because weather data are not available for some cities. Specifications with city-week fixed effects also drop city-weeks with only one observation.

In addition, given the city-week fixed-effects included in our specifications for Columns (3) and (6), we are somewhat limiting the level of identifying variation available. Reassuringly, though, we see that there are, on average, 26 observations in each city-week (ranging from 2 to 73 observations per city-week).

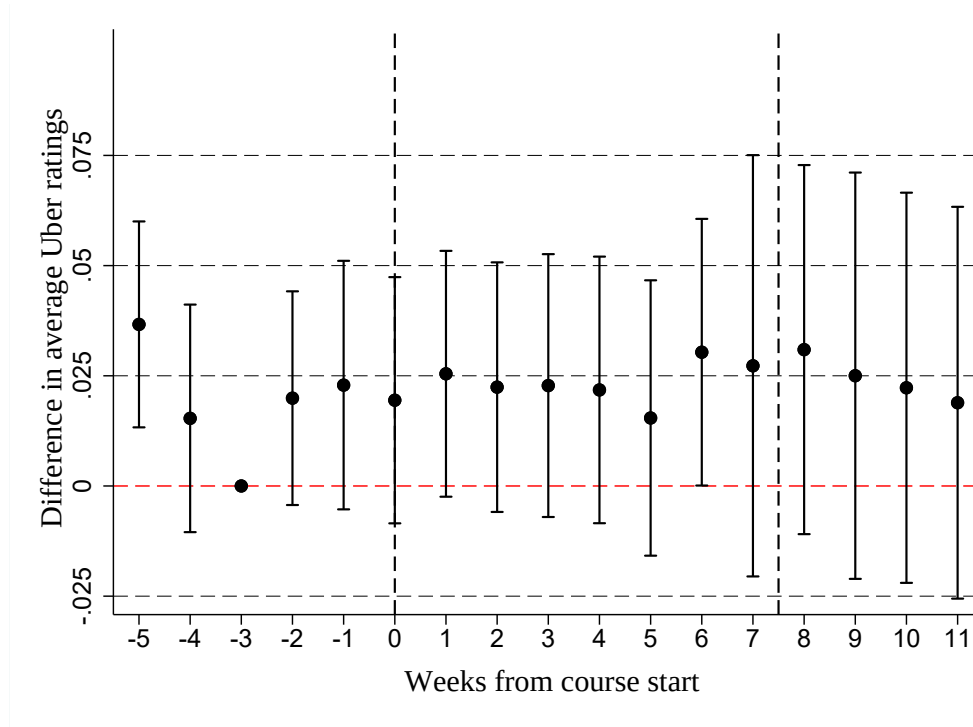
Turning to labor market conditions, the estimates in Table 6 (which should be interpreted with some caution per the discussion in Section 5.1) show that a \$1 increase on average hourly pay increases average weekly Uber hours by 0.122 hours, or about 7.3 minutes, and completed trips by 0.35 trips per week.²⁹ If we consider that the average *active* hourly pay of drivers that do not attend ASU is \$33.60, and that the average weekly driving time among Uber-students is 18.24 hours, then a 10% increase in average hourly pay leads to an increase in driving time (trips) of roughly 2.3% (3.2%). Caldwell and Oehlsen (2023) finds that the Frisch elasticity for male Uber drivers (our sample is overwhelmingly male) is 0.4, which is higher than the elasticity we estimate. However, two reasons could explain the differences in findings. First, Caldwell and Oehlsen (2023) estimates labor supply elasticities using random variation generated by an experiment that allowed drivers to work for one week with 10-50% higher earnings per trip. Because the experiment explicitly highlighted the potential gains from additional driving, it is natural that drivers may have been more responsive in that setting. Second, our sample of drivers tends to have a stronger attachment to work (i.e., they consistently drive more hours per week than the average Uber driver), likely making them less responsive to changes in market hourly pay.

Our data also allows us to analyze whether the quality of the service that drivers offer is affected by college activities. To this end, we performed a similar descriptive event-time analysis as in Section 3.1 but with the ratings the drivers received from their passengers and the share of income from tips as dependent variables. Figure 3 shows no negative impact on ratings, while Appendix Figure H.3 shows a small drop in the fraction of earnings coming from tips, but the economic impact is extremely small. Thus, we conclude that college activities have no meaningful impact on the quality of service.

Overall, our findings indicate that the “short run” opportunity cost of coursework in enrolled weeks is modest: approximately \$180 per month relative to average monthly Uber earnings of about \$1,900 among students taking ASU classes. At the same time, we do not observe any meaningful negative impact of college activities on the quality of work performed. Extrapolating these results out over the entirety of a bachelor’s degree worth of

²⁹The top sub-figures of Appendix Figure H.2 present the distribution of the labor supply elasticity with respect to coursework and earnings (i.e., calculated based on the average coursework and market-level hourly earnings of each Uber-student).

Figure 3: Variation in driver ratings during eight-week courses



Notes.—OLS coefficient estimates from a version of Equation (1) where the dependent variable is average driver rating and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

courses, these results suggest that an average student in the program would forgo roughly \$7,560 to receive a degree.³⁰ In Section 9.2, we combine this opportunity-cost estimate with evidence on longer-run credit accumulation and graduation rates to assess the wage return required for participation in the program to be financially worthwhile.

6.2 Labor Market Conditions and Study Effort

Table 7 presents results from estimating the effects of labor market conditions and college course demand on Uber-students' Canvas hours. Evidence on how variation in weekly earnings impacts study effort is scarce. Thus, the estimates in Table 7 are unique in that they

³⁰We calculate the total opportunity cost of receiving a degree by assuming that students need to complete 40 courses to graduate (a minimum of 120 credits are required and over 85 percent of the courses taken in our sample are worth 3 credits). Multiplying 40 courses by the product of average number of peer study hours per course (11.47 hours) and our estimate of work hours forgone per peer Canvas hour (-0.68) yields a total reduction of roughly 312 work hours, which we then scale by average hourly earnings while taking courses (\$24.26).

quantify the extent to which Uber-students' study time responds to short-run changes in earnings opportunities. Column (1) shows that study effort is only modestly responsive to transitory changes in labor market conditions. A \$1 increase in average hourly pay decreases study time by less than a minute per course-week. Equivalently, a 10% increase in average hourly pay reduces study time for the average Uber-student by about 1.7% per course. This finding suggests that study time is not highly sensitive to changes in labor market conditions.³¹ However, this result could be specific to our context, given that flexibility may allow drivers to spend more time on the Uber platform when earnings are high without meaningfully decreasing their study time.

Column (1) also shows that average peer Canvas hours have a nearly one-to-one relationship with Uber-student study hours, suggesting that our measure of course demand effectively captures the variation in college work across weeks.³² By contrast, we find little evidence of crowd-out across courses, given that an increase in coursework demand in other courses does not meaningfully reduce study time in course c . Column (2) presents a related specification that uses the number of assignments graded as an alternative measure of course demand. The results imply that an additional assignment in course c increases weekly average study hours in course c by about 2.8 minutes (0.046 hours), or by around 3% relative to the mean.

Column (3) presents a placebo test designed to assess whether market-level hourly pay is proxying for unobserved shocks that also affect online study behavior. For example, suppose weather conditions (which we control for) simultaneously affect study hours and earnings. In that case, we should observe a mechanical correlation between Uber's market hourly pay and study hours for students who do not drive with Uber. To this end, we repeat the specification in Column (1) but on the sample of students attending ASU Online classes but not participating in the Uber-ASU program. Reassuringly, we find a coefficient of zero for

³¹A potential concern for external validity is that Uber students may be positively selected for their time management, making our results limited to our sample. However, Appendix C.2 shows no evidence of smoother within-semester study patterns among Uber-students relative to their classmates; if anything, their weekly Canvas hours are more variable within the same course offering. This suggests that our findings are unlikely to be driven by a subgroup with unusually strong time-management skills.

³²The bottom subfigures of Appendix Figure H.2 present the distribution of course effort elasticity with respect to coursework and earnings, calculated based on the average coursework and earnings of each Uber-student.

Table 7: Determinants of study effort

	Study Hours in c (1)	Study Hours in c (2)	Placebo Test (3)
Avg. study hours in c , peers	1.134*** (0.027)	-	1.022*** (0.018)
Sum of avg. study hours in $-c$, peers	-0.009 (0.006)	-	-0.007*** (0.002)
Assignments graded in c	-	0.046*** (0.015)	-
Sum of assignments graded in $-c$	-	-0.012** (0.005)	-
Market level active hourly pay, cur- rently elig.	-0.008** (0.004)	-0.029*** (0.004)	-0.001 (0.001)
Observations	55293	55293	555752
Individual Fixed Effects	✓	✓	✓
Classroom Fixed Effects	✓	✓	✓
Month-Year Fixed effects	✓	✓	✓
Weather controls	✓	✓	✓
Mean dep. var.	1.58	1.58	-
Avg. study hours in c , peers	1.27	-	-
Sum avg. study hours in $-c$, peers	1.55	-	-
Assignments graded in c	-	2.21	-
Sum assignments graded in $-c$	-	2.78	-
Market level active hourly pay, peers	33.6	33.6	-
Elasticity w.r.t. peer coursework	0.912	0.064	
Elasticity w.r.t. peer earnings	-0.170	-0.617	

Note: The dependent variable in each column is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . The variable “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . “Assignments graded in c ” and “Sum of assignments graded in $-c$ ” denote the number of assignments graded in course, c , and all other courses, $-c$ for a student in a given week. The results in Column (2) are from a 2SLS specification where “Assignments graded in c ” and “Sum of assignments graded in $-c$ ” are instrumented with the average graded assignments for non-Uber peers in c and $-c$, respectively. “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Elasticities are calculated by multiplying the relevant coefficient estimate by the ratio of mean peer coursework or market level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

these students.³³

Taken together, Tables 6 and 7 quantify the short-run trade-offs that Uber-students face in terms of work-study time allocation in this flexible environment. While we emphasize these trade-offs in terms of elasticities, one might be inclined to instead focus on the implied level changes in work and study hours. However, baseline study time is much lower than baseline work time, and both time-use outcomes are bounded below by zero with substantial mass near that boundary. As a result, hour-for-hour comparisons can be difficult to interpret and may implicitly extrapolate outside the feasible support. We therefore emphasize elasticities, which place the two margins on a common, scale-free metric and summarize substitution around observed baselines.

Moreover, because our identifying variation is week-to-week, both the elasticity and level estimates should be interpreted as within-semester, intensive-margin allocation effects. They capture how current work changes when coursework becomes more demanding and how current study effort changes when labor market opportunities improve. In this sense, the estimates are informative about short-run strategic scheduling and marginal substitution, but they should not be interpreted as the cumulative year-level crowd-out of work and schooling activities.³⁴

Appendix Table G.9 further shows that the study-effort estimates are robust to including leads and lags of the independent variables. We return to the interpretation of these dynamic specifications in the next subsection, which examines whether our contemporaneous estimates mask anticipatory responses. Appendix Table G.10 replicates the analysis in Table

³³As an additional robustness check, we re-estimate Table 7 specification with week-year fixed effects. This is feasible because identification can still come from differences across geographic markets. However, because our measure of market-level active hourly pay is constructed at the city-week level, these more saturated time controls absorb most of the common time-series variation in that regressor. Consistent with this mechanical implication, the coefficient on market-level hourly pay becomes close to zero and statistically insignificant, while the estimated effects of course demand on study time remain very similar to our baseline results (Appendix Table G.8). We therefore retain the month-year fixed-effects specification in Table 7 as our baseline, and view the week-year fixed-effects results as further evidence that Uber-students' study effort is not meaningfully driven by short-run labor market shocks.

³⁴While our main estimates should be interpreted as short-run, intensive-margin responses, Appendix C.1 provides complementary descriptive comparisons of weekly Uber outcomes during enrolled and non-enrolled weeks. These comparisons do not provide causal estimates of the extensive-margin effect of course enrollment on work, but they provide a useful point of comparison for our short-run estimates by showing how Uber outcomes differ in weeks when drivers are enrolled in ASU courses relative to weeks when they are not. Reassuringly, these descriptive patterns are broadly consistent with the magnitude of our main short-run opportunity cost estimates.

7 using weekly clicks and logins as alternative measures of study effort, with qualitatively similar results. Finally, Appendix D combines the estimates from Tables 6 and 7 with the analytical framework to decompose the marginal costs of work and study effort. The exercise suggests that the cost of working is largely independent of concurrent coursework, while the cost of studying is more strongly amplified by simultaneously working.

6.3 Robustness to Anticipatory Responses

One potential concern with our contemporaneous weekly estimates is that coursework demands may be partly predictable over the academic term. If students anticipate upcoming spikes in coursework—for example, around midterms, finals, or major assignments—they may adjust their work schedules in advance rather than in the same week. In that case, our baseline estimates could understate the total labor-supply response to coursework demands.

To assess this possibility, we augment our baseline weekly labor-supply specifications with a one-week lead of our course-demand proxy and examine whether future workload predicts current work behavior. Appendix Table G.11 reports the results. If students systematically reduce labor supply in advance of predictable increases in coursework, future course demand should negatively predict current driving hours. Instead, we find no evidence of such anticipatory adjustment. The estimated relationship between next week’s course demand and current driving hours is small and statistically indistinguishable from zero. These results suggest that the modest elasticity we estimate is unlikely to be driven by systematic one-week-ahead anticipation of coursework shocks. Rather, the evidence is more consistent with adjustment occurring contemporaneously through within-week reallocation and scheduling responses.

We find similar results when examining potential anticipatory responses of study behavior to future course demands and labor-market conditions. Appendix Table G.12 shows no evidence that students adjust current study hours when course demands or wages are higher in the following week. Moreover, favorable labor-market conditions in the previous week do not lead students to increase study time in the current week. If anything, the estimated relationship is weakly negative. Overall, these results support the interpretation of our main estimates as capturing contemporaneous, within-week adjustments of work and study effort.

6.4 Heterogeneity in Work-Study Trade-Offs

Next, we examine whether changes in coursework loads and labor market conditions lead to differential responses based on Uber-students' observable characteristics. In particular, we consider i) full-time or part-time driver status, since time constraints may be more binding for full-time drivers; ii) field of study, since certain majors may be more demanding than others; iii) gender, as prior literature demonstrates that labor supply choices may differ by the gender of the driver (Cook et al., 2021; Caldwell and Oehlsen, 2023); iv) financial need status, since labor supply may be more inelastic to changes in coursework demand for this group; v) incoming academic performance (above- vs. below-median incoming GPA), as certain students may be more responsive to coursework demands; and vi) age (under 35 vs. 35 or older), given that schedule flexibility may differ by age (because of family responsibilities etc.).

Panels A and B of Table 8 examine heterogeneity in the main work and study effort responses. Panel A re-estimates Equation (6), interacting coursework demand with observable student characteristics. Panel B re-estimates Equation (7), interacting market-level hourly pay with the same characteristics.

Column (1) of Panel A shows that full-time drivers reduce labor supply more strongly in response to coursework demand than part-time drivers. We define full-time drivers as those who average at least 25 hours of driving per week when not enrolled in ASU courses. This pattern is consistent with full-time drivers facing tighter time constraints and having less scope to absorb coursework demands without reducing work hours. An alternative explanation is that program eligibility requirements may be more binding for part-time drivers, making their work hours less responsive. However, as discussed in footnote 6, the average work activity of part-time drivers is substantially above the threshold required to maintain eligibility.

Column (2) of Panel A shows that STEM students also reduce work hours more strongly in response to coursework demand. In particular, STEM students' labor supply is roughly 79% more responsive to changes in college activities than that of non-STEM students (-0.645 vs -0.360). This pattern is consistent with STEM courses placing greater demands on stu-

Table 8: Heterogeneity in the determinants of work and study hours

	Full-time driver (1)	STEM (2)	Female (3)	High-need (4)	High incoming GPA (5)	Age $\geq$ 35 (6)
<i>Panel A: Work hours</i>						
Sum avg. study hours all c, peers	-0.381*** (0.090)	-0.360*** (0.092)	-0.497*** (0.094)	-0.554*** (0.132)	-0.465*** (0.115)	-0.489*** (0.118)
Sum avg. study hours all c, peers $\times$ subgroup	-0.237* (0.137)	-0.285** (0.136)	0.166 (0.157)	0.079 (0.144)	0.027 (0.134)	-0.005 (0.132)
Observations	27404	26636	26690	27388	23902	27404
<i>Panel B: Study hours</i>						
Market level active hourly pay, currently elig.	-0.010** (0.004)	-0.007 (0.005)	-0.007* (0.004)	-0.011* (0.006)	-0.007 (0.005)	-0.010* (0.005)
Market level active hourly pay, currently elig. $\times$ subgroup	0.006 (0.006)	-0.001 (0.006)	-0.015 (0.010)	0.003 (0.007)	-0.003 (0.006)	0.004 (0.006)
Observations	55293	53731	53975	55266	48263	55293

Note: This table only reports the relevant coefficients, though the specifications follow equations (6) and (7) but with the inclusion of the relevant interactions with the different subgroups. “Full-time drivers” are those that average at least 25 hours of driving per week when not enrolled in ASU classes. “High-need” indicates that Uber-students are either “High” or “Very High” need based on their FAFSA records, and “High incoming GPA” indicates that the student’s incoming GPA was at least 2.72 (the sample median). Appendix Table G.13 shows all the specification coefficients. For Panel A—the dependent variable in each column is hours working on the Uber platform. “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership but not enrolled. “Sum of avg. study hours across all c, peers” denotes the sum of the average hours per week that peers spent on Canvas across all courses. For Panel B—the dependent variable in each column is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . The variable “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . All specifications include multiple sets of fixed effects, second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. In Panel A, standard errors are clustered at the individual level. In Panel B, standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

dents’ time.³⁵ Finally, we do not find statistically significant differences in the responsiveness of work hours by gender, financial need, incoming GPA, or age.

Panel B examines heterogeneity in the study-effort response to market-level hourly pay. Overall, we find little systematic heterogeneity in how labor market conditions affect Canvas hours. The interaction terms are generally small and statistically insignificant, suggesting that the modest study-effort response to labor market conditions is broadly similar across observable student groups.

³⁵Our estimates indicate that the return to an individual hour of time spent on Canvas is lower in terms of course grades for STEM students. Specifically, we find that the return to an hour of time on Canvas is only 70% as high as compared to non-STEM students. This suggests that STEM majors need to spend relatively more time on Canvas in response to their course activities. In fact, we find that, on average, Uber and non-Uber students in STEM majors spend over two hours more on Canvas for each course.

Overall, Table 8 indicates that short-run work-study trade-offs are broadly similar across students with a range of observable characteristics. The clearest exceptions are full-time drivers and STEM students, for whom the short-run opportunity cost of coursework appears somewhat larger.

7 The Role of Flexibility

The working-learning environment we study offers substantial flexibility in both work schedules and coursework. It is therefore useful to assess whether Uber-students actively adjust the timing of their work when course demands increase. The results in Section 6 indicate that most Uber-students face relatively modest short-run trade-offs between school and work. In this section, we examine whether within-week re-timing of work is one margin through which students accommodate those changing demands.

7.1 Do Drivers Adjust Working Hours in Response to College Demands?

To investigate whether Uber-students actively re-time their work schedules when course demands change, we first construct a dissimilarity index that captures deviations in the driving schedules of Uber-students from their typical driving patterns when not enrolled in classes. In particular, our data allows us to observe how drivers distribute their time on the Uber platform across 28 six-hour periods in each week; for example, the first period is Monday morning from 12:00 AM to 6:00 AM and the second is Monday from 6:00 AM to 12:00 PM. We use these data to construct an index that reflects overall deviations from a driver’s usual driving schedule when not enrolled at ASU. In a given week, t , Uber-student i ’s total dissimilarity index is equal to:

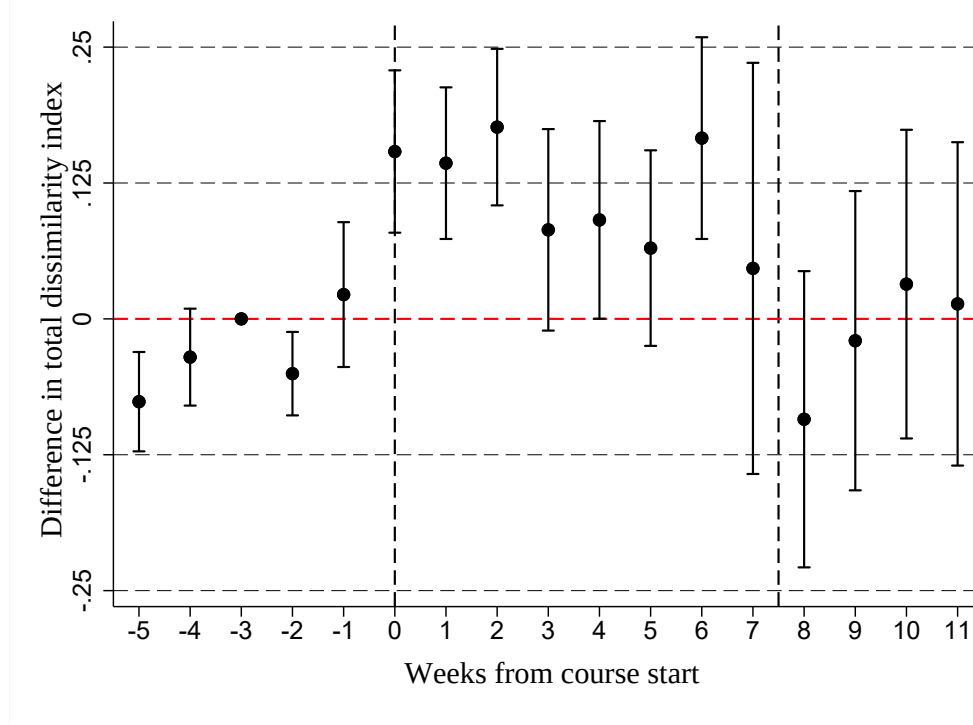
$$D_{i,t} = \frac{1}{2} \sum_{p=1}^{28} \left| \frac{\bar{m}_{i,p}}{\bar{M}_i} - \frac{m_{i,p,t}}{M_{i,t}} \right|, \quad (10)$$

where p refers to one of the 28 six-hour periods, $m_{i,p,t}$ is the total minutes driven in period p in week t , $M_{i,t}$ is the total number of minutes driven in week t , $\bar{m}_{i,p}$ is the average minutes driven across all p periods in non-ASU weeks (i.e., when not enrolled in classes), and $\bar{M}_i$ is the average number of minutes driven across non-ASU weeks. This measure compares the within-week allocation of driving time in week t to the driver’s typical non-ASU driving schedule. Because the index is based on the share of weekly driving minutes allocated to each of the 28 periods, it intentionally differences out proportional changes in total weekly hours. Thus, if a driver simply reduces driving uniformly across the week while preserving the same relative timing pattern, the index does not change. In that sense, the index isolates within-week re-timing of work rather than reductions in total driving hours. Interpreted as a total-variation distance from the driver’s typical non-ASU schedule, larger values indicate more active reshuffling of driving time across days and time periods. For example, $D_{i,t} = 0$ implies no change relative to the typical non-ASU schedule, whereas a value of 1 implies a fully dissimilar schedule. For ease of interpretation, we standardize the index in the empirical analysis.

Figure 4 presents results from a specification analogous to the event-time exercises previously presented but using the (standardized) dissimilarity index described above as the dependent variable. Because the index is standardized, each coefficient can be interpreted in standard-deviation units relative to the omitted event week (Week -3). Positive coefficients indicate that students’ driving schedules deviate more from their typical non-enrollment schedules. The estimates show that driving schedules become more dissimilar once courses begin. In the first week of an eight-week course, the dissimilarity index increases by roughly 14% of a standard deviation. The increase is largest near the beginning and end of the course, when Canvas engagement also tends to be highest, and smaller during the middle of the course. These patterns suggest that students take advantage of within-week scheduling flexibility to adjust the timing of work when course demands are highest.

The magnitude of the change is economically meaningful. Appendix E shows that the variation in the dissimilarity index in Figure 4 is similar in magnitude to randomly reallocating roughly 20% of a driver’s non-course driving schedule across within-week time periods. We therefore interpret the results as evidence that students actively re-time work during

Figure 4: Variation within week driving flexibility during eight-week courses



Notes.—OLS coefficient estimates from a version of Equation (1) where the dependent variable is the standardized dissimilarity index and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

enrolled weeks, rather than simply reducing hours uniformly across the week.³⁶

Appendix C.2 provides complementary evidence that Uber-students also make use of flexibility in their coursework. Relative to other ASU Online students in the same course offerings, Uber-students exhibit greater week-to-week variability in Canvas hours. Taken together with the results above, the patterns are consistent with Uber-students using flexibility on both margins by re-timing work within the week and allowing study effort to vary across weeks.

³⁶These event-time estimates translate to an average treatment effect across weeks that can be summarized as follows: when the average study time of their peers increases by one hour per week, on average, drivers deviate from their typical schedule by roughly 4.1% of a standard deviation. Moreover, Appendix Figure H.4 presents similar results where we construct the dissimilarity index specifically for typical peak hours periods (i.e., Friday and Saturday afternoons and nights). The results from the figure suggest that these driving behaviors during those hours are particularly likely to adjust during course weeks.

Table 9: Average Stated Likelihood of Enrollment by Cost and Flexibility Scenario

	Uber-students (1)	Class Peers (2)	Diff. P-values (3)
Enrollment likelihood with cost = \$0			
Fixed work schedule x In-person Classes	30.51	32.80	0.40
Fixed work schedule x Online classes	73.12	84.92	0.00
Flexible work schedule x In-person classes	47.62	55.38	0.01
Flexible work schedule x Online classes	88.95	94.22	0.00
Enrollment likelihood with cost = \$6,000			
Fixed work schedule x In-person classes	17.50	26.10	0.00
Fixed work schedule x Online classes	33.18	70.13	0.00
Flexible work schedule x In-person classes	25.51	42.82	0.00
Flexible work schedule x Online classes	43.74	78.82	0.00
Observations	243	344	

Note: “Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to students that are enrolled in the same classes as the Uber students. Column “Diff. P-values” reports the p-values corresponding to the differences between columns (1) and (2).

7.2 Assessing Students’ Valuation of Flexible Schedules

Next, we focus on uncovering students’ valuation for flexibility in learning and working schedules. To this end, our survey elicited students’ likelihood of pursuing higher education under varying cost and flexibility scenarios. More specifically, the options were differentiated by three key factors: (1) the nature of the work schedule (i.e., flexible or fixed), (2) the modality of obtaining college credits (i.e., online or in-person), and (3) the annual cost of education (\$0 or \$6,000, with the latter reflecting approximately the regular yearly fees for an ASU working student). In the survey, students were asked the following:

“Consider the case where your education cost is [SPECIFIC VALUE HERE] per academic year. What is the percent chance (or chances out of 100) that you will enroll in the bachelor’s program in each of the following four cases? When answering these questions, assume that the number of hours you work and your earnings are the same across scenarios:

a) your work schedule is fixed and do not have flexibility AND you can take college credits toward a bachelor’s degree online.

b) your work schedule is fixed and do not have flexibility AND you can take college credits towards a bachelor's degree in person only.

c) your work schedule is flexible AND you can take college credits toward a bachelor's degree online.

d) your work schedule is flexible AND you can take college credits towards a bachelor's degree in person only.”

Each student therefore provided eight responses: four flexibility and modality scenarios at each of two education costs. Table 9 summarizes the average likelihood of enrollment in each scenario, separately for Uber-students and class peers. Uber-students, often from less privileged backgrounds (see Table 1), showed a greater sensitivity to education costs than their classmates, with enrollment probabilities decreasing by about 50% when costs rose from \$0 to \$6,000, irrespective of the learning and working formats. Students also report a higher likelihood of enrollment in scenarios that combine online learning with flexible work schedules. For instance, moving from the fully rigid environment—fixed work schedules and in-person classes—to the fully flexible environment—flexible work schedules and online classes—substantially increases the reported likelihood of enrollment. Therefore, these findings strongly suggest that this sample of students places considerable value on flexible formats when considering college enrollment.

The table also suggests that flexibility in learning modality is especially important. For both Uber-students and their class peers, the mean likelihood of enrollment increases by roughly a factor of two or greater going from in-person classes to online classes (when keeping the work schedule fixed).

Appendix F uses these survey responses to recover model-based measures of willingness to pay (WTP), following the expected-utility framework in Aucejo et al. (2023). The structural exercise yields two main findings. First, estimated WTP for flexible learning is two to three times as large as estimated WTP for flexible work schedules. Second, because Uber-students appear more sensitive to education costs, their estimated WTP for flexible study or work arrangements is lower than that of their class peers.

8 Effort Measures and Academic Performance

We next examine how work and study effort affect students' weekly and overall course-level performance. Panel A of Table 10 presents results from estimating different versions of equation (8), which focuses on course weekly performance, as measured by the share of points earned in the week (i.e., $\frac{\text{points earned}}{\text{points possible}} \times 100$). Although this normalization removes the mechanical relationship between points at stake and points earned, common course-week difficulty may still affect both peer study time and Uber-students' normalized performance; we therefore interpret the estimates presented in Panel A as supportive evidence on the performance consequences of study-time adjustments. Column (1) considers total Uber (work) hours, while Column (2) refers to a specification that accounts for Canvas (study) hours. Both columns present results from specifications that instrument Uber-students' weekly hours on Uber and Canvas with the average market-level pay for eligible drivers and average hours on Canvas of class peers (excluding other drivers in the ASU-Uber program), respectively. The results in Panel A suggest that spending more hours on the Uber platform may have a small negative (but economically meaningless) effect on weekly course performance: each additional hour driving is associated with a (small) decline in the share of points earned of about 0.6 points on a 0-100 scale. As expected, Column (2) shows that an additional hour on Canvas increases the point share attained. Though seemingly modest, we later show that this effect appears large enough to meaningfully affect aggregate course performance.

Next, we explore how aggregate effort throughout the term translates to overall course performance. Panel B of Table 10 shows results corresponding to specifications that include course (e.g., ECN 101) and individual-by-term fixed effects, as described in Section 5.3. As with Panel A of the table, Column (1) considers total Uber hours, and Column (2) presents results from a specification that accounts for Canvas hours. In Column (1), we instrument an individual's total driving hours throughout the course with the average market-level hourly pay for drivers who were eligible for the ASU-Uber program but did not enroll, averaged across all course weeks. The specification presented in Column (2) instruments a student's Canvas hours with average total hours on Canvas of class peers. For robustness, Appendix

Table 10: The effect of work and study effort on course outcomes

Panel A: Weekly performance		
	(1) Point Share	(2) Point Share
Work hours	-0.614** (0.293)	-
Study hours	-	1.020*** (0.369)
Obs.	55293	55293
First-stage F-stat.	39.57	2058.84
Individual Fixed Effects	✓	✓
Classroom Fixed Effects	✓	✓
Month-Year Fixed Effects	✓	✓
Weather Controls	✓	✓
Avg. dep. var.	86.38	
Avg. study hours	1.58	
Avg. work hours	17.71	
Panel B: Final course grades		
	(1) Course Grade Points	(2) Course Grade Points
Total work hours	0.001 (0.003)	-
Total study hours	-	0.078*** (0.016)
Obs.	4259	4259
First-stage F-statistic	23.76	73.51
Adjusted R ²	0.61	0.64
Course Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.	2.54	
Avg. total hours on Canvas	13.50	
Avg. total Uber hours	142.16	

Notes: Panel A—The dependent variable, “Point share”, is on a scale of 0 to 100. “Work hours” refers to the number of hours spent on the Uber platform in a given week, and “Study hours” denotes the number of hours a student was connected to Canvas for a given class each week. Each regression model includes individual, classroom, and month-year fixed effects. The model in Column (1) is estimated via 2SLS where “Work hours” is instrumented with the weekly average market-level pay for Uber drivers eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where “Study hours” is instrumented with the average Canvas hours for non-Uber peers in the course for a given week. Panel B—The variable “Total work hours” denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. “Total study hours” refers to the sum of individual Uber-student’s own Canvas hours throughout a given course. Each regression model includes course (e.g., ECN 101), and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the Covid-19 pandemic. The model in Column (1) is estimated via 2SLS where “Total work hours” is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where “Total study hours” is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the course level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Tables G.14 and G.15 present results of similar specifications that examine how work and study time affect course completion and pass rates.

Three main takeaways emerge from the results in Panel B. First, we find no evidence that additional driving hours reduce course grades. Second, based on the estimates from the IV specifications in Panel B, we find that a 10% increase in the average total Canvas hours (i.e., 1.35 hours) increases course grades by 0.11 points (i.e., 1.35×0.078), approximately one-tenth of a grade point, which is consistent with the positive effects found in Column (2) of Panel A. This result indicates that Canvas hours are strongly associated with course performance, suggesting that, though imperfect, our measure of study hours is highly relevant. Finally, using the estimates in Panel B, it is possible to provide a back-of-the-envelope calculation to quantify how changes in labor market conditions (i.e., market hourly pay) impact students' academic performance through a reduction in study hours. Combining the estimates from Tables 7 and 10, we find that a 10% increase in average market-level hourly pay will lead to a small decrease in course grade performance, roughly 0.016 grade points.³⁷

Overall, the findings reported in Panels A and B indicate that additional Uber work effort is not associated with economically meaningful declines in academic performance in this setting. This pattern is consistent with the evidence in previous sections showing limited short-run displacement between work and coursework, and suggests that flexible work and study arrangements may allow Uber-students to adjust their schedules in ways that largely preserve course performance.

9 Perceived Returns and Degree Progress in the ASU-Uber Program

In this section, we complement the high-frequency evidence on work-study trade-offs with evidence on students' perceived returns to degree completion and realized progress in the

³⁷The point estimates from Table 7, imply that a 10% (\$3.36) increase in average market-level hourly pay of peers would lead to a reduction in weekly study hours by 0.027 hours (i.e., $\$3.36 \times -0.008$). If we multiply this number by 7.5 (most courses are either 7 or 8 weeks long), then the decrease in the total number of study hours for a given course is around 0.202 hours. Finally, if we use the estimate from Column 2 of Table 10 (Panel B), jointly with 0.202, we obtain the 0.016 point decrease in the course grade.

ASU-Uber program.

9.1 Students' Perceived Returns to Program Participation

Our survey collected students' perceptions about their expected graduation rates and future income under various scenarios, allowing us to characterize the ex-ante value students attribute to the program. These subjective beliefs—whether or not they are ultimately realized—are central for understanding participants' decisions to enroll in the program.

Appendix Table G.16 reveals that students anticipate an average graduation rate exceeding 90%. We also inquired about the expected graduation rate of transfer students had they remained at their previous institutions. Students' responses indicate a marked improvement in anticipated graduation rates due to transferring, with an increase of about 25 percentage points. As to whether these expectations will be ex-post realized is a separate question, but ex-ante these students are quite serious about completing their programs. Finally, the bottom panel of Table G.16 presents the ratio of the students' projected annual income in six years, conditional on obtaining a bachelor's degree or not (i.e., dropping out of the program). On average, students foresee a 90% return from earning a bachelor's degree, highlighting the significant perceived value they attach to completing the program.

It is admittedly too early to determine whether students will ultimately realize such substantial returns to program participation; in addition, we are unable to track the program participants. However, it is crucial to note that the online programs examined in this paper differ markedly from many conventional short online post-secondary offerings, as we examine bachelor's degree programs offered by a large public university that does not distinguish between online and in-person degrees.³⁸ Moreover, recent employer surveys further suggest growing recognition of online degrees as equivalent to traditional, in-person qualifications. For instance, the National Association of Colleges and Employers found that 87.4% of employers who track degree modalities have hired graduates with online degrees, offering them

³⁸Appendix Table G.17 shows that 96% of Uber students considered ASU education to be either better (50%) or comparable (46%) to their prior institutions' quality, with similar perceptions also reported by their class peers. While these shares should be interpreted with caution because they are conditional on students currently enrolled at ASU Online, they suggest that Uber-students are not trading off college quality for more flexible, free education.

the same starting salaries as those awarded to candidates with traditional degrees.³⁹ These developments suggest that online degrees are increasingly viewed as credible credentials, which helps contextualize why students in our sample report substantial expected returns to completing an ASU Online degree.

9.2 Graduation Rates and Credit Accumulation Among the Early ASU-Uber Cohorts

The survey evidence above captures participants' perceived returns to degree completion and their expectations about educational progress. We next complement these subjective measures with administrative evidence on credit accumulation and graduation. Although data limitations prevent us from tracking these outcomes for the drivers in our analytic sample beyond May 2021, we were able to obtain aggregate graduation and credit-accumulation statistics for the first four cohorts of ASU-Uber program participants, from which our analytic sample is derived.

Appendix Table G.18 provides a summary of both persistence and degree progress among the first four ASU-Uber entry cohorts. As of Fall 2025, 27.2% of participants in these cohorts had graduated, while the cohort-size-weighted five-year graduation rate is 25.7%. This gap suggests that a meaningful share of participants complete on a longer timeline, which is unsurprising given the flexibility of the program and the demands faced by working students.⁴⁰ At the same time, the average entrant accumulated 81 ASU credits, or roughly two-thirds of the 120 credits required for a bachelor's degree. These figures indicate that, although many participants do not complete the program within five years, the typical entrant nevertheless makes substantial progress toward a degree, which is important for

³⁹It is also important to mention that the returns to a college degree remain high even for marginal students (Zimmerman, 2014). While Deming et al. (2016) finds that degrees from online for-profit institutions lead to lower employer callback rates than comparable degrees from nonselective public schools, our context differs notably in two respects. First, ASU is a widely recognized public institution, well regarded for its online programs. Second, ASU transcripts do not specify whether degrees were earned online or in-person, potentially mitigating modality-related factors.

⁴⁰As a benchmark, the six-year bachelor's degree graduation rate for full-time, first-time bachelor's degree-seeking undergraduate students at ASU Digital Immersion is 24% (National Center for Education Statistics, 2023). This comparison suggests that Uber participants enrolled at ASU may have graduation rates somewhat higher than the relevant online undergraduate benchmark.

interpreting both the costs and potential benefits of participation.

Relying on this additional information on credit accumulation and graduation rates, we can calculate how long it would take the average entrant to offset the opportunity cost of participation through higher post-graduation earnings. Scaling our estimated full-degree opportunity cost of \$7,560 in proportion to 81 accumulated credits implies expected forgone earnings of roughly \$5,100 per entrant. If we conservatively assume that earnings gains accrue only to participants who graduate, then expected benefits must likewise be scaled by the observed graduation probability of 27.2%. Assuming an illustrative 15% post-graduation wage gain—which falls close to the middle of estimates in the literature of the returns to college *enrollment* (Mountjoy, 2026; Zimmerman, 2014)—and annual Uber earnings of approximately \$24,000, the average entrant would offset expected opportunity costs after roughly five years. This payback calculation provides a simple entrant-level benchmark for evaluating whether the program’s longer-run earnings gains are large enough to justify its short-run opportunity costs.⁴¹

Taken together, the evidence in this section suggests that the ASU-Uber partnership addresses margins that are important for working adults considering higher education. Participants report high expected returns to completing an online degree, and administrative records show that early program cohorts make meaningful degree progress: the average entrant has accumulated roughly two-thirds of the credits required for a bachelor’s degree, and more than one-quarter had graduated by Fall 2025.

10 Conclusion

This is the first study to quantify how college activities and labor market conditions affect labor supply and study effort at a high frequency. Our findings indicate that short-run work-study trade-offs are modest in this highly flexible setting. In particular, we find that average college activities generate an average short-run opportunity cost of about \$41 per week, while

⁴¹We present this as an undiscounted payback benchmark rather than a formal present-value calculation because discounting would require additional assumptions about the timing of both costs and benefits. In particular, students accumulate credits over different horizons, and post-graduation earnings gains may begin at different points depending on time to degree.

a 10% increase in average hourly pay decreases study time for the average Uber-student by only about 1.7%. These estimates are consistent with additional evidence that Uber-students actively re-time their driving behavior when course demands increase. Importantly, we also find limited evidence that additional work effort reduces academic performance or that college activities reduce the quality of workplace performance.

Survey evidence further indicates that students in our sample place a substantial value on flexibility in learning environments and report meaningful expected returns to completing an online degree. We interpret these patterns as suggesting that flexibility may matter for participation in higher education, while recognizing that our data do not allow us to identify the program’s causal effect on college demand or to conduct a full welfare analysis.

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Online Appendix: Supplementary Material

A Additional Data Description

A.1 Sample Restrictions

The primary way in which we restrict our data to construct our estimation sample is to limit the weeks included in our analysis. In particular, we drop all weeks between March 16, 2020, and January 4, 2021. The motivation for doing this is that the COVID-19 pandemic and related stay-at-home orders led to a massive disruption in Uber driver activity. We include the Spring term of 2021 in our analysis because, by that point, Uber driving behaviors roughly returned to pre-pandemic levels, and a large share of our sample comes from that term. In addition to dropping these observations, we also exclude “Dynamic” courses from our estimating sample. These courses are often only one week long and make up a very small share of our Uber-student sample. Besides these restrictions, we do not place any other explicit filters on our estimating sample.

For our estimation of Equation (1) (i.e., analysis of data variation), we also make additional sample restrictions. Specifically, we exclude individuals who appear to have “dropped out” during the course. We determine likely dropouts based on their final course grades (e.g., incomplete or withdrawals) and their activity. Specifically, if a student receives a failing grade, an incomplete, or a withdrawal *and* has a spell of inactivity for at least the final three weeks of the course, we consider them a likely dropout. Though excluding these observations from our main results has little effect, they do matter in the estimation of our data variation analysis since dropouts meaningfully attenuate our estimates in weeks after they drop out (since their driving hours are relatively high and study hours relatively low due to dropping out).⁴²

⁴²Appendix Tables G.19 and G.20, present results where we re-estimate our main specifications but limit the sample by excluding likely dropouts.

A.2 Select Variable Definitions

In this subsection, we provide an overview of key variables used in our analysis. In terms of relevant measures of study effort, we directly observe the amount of time that students are connected to a course’s Canvas page, as well as the number of logins and clicks that a student generates each week. To construct the measures of course demand in a given week, we calculate the average amount of time spent on Canvas by all course peers (i.e., non-Uber program participants).

For work effort/labor supply measures, we observe how much time a driver is connected to the Uber application and how many weekly trips they complete. Note that, at the driver level, we cannot distinguish between active and inactive time, where active time refers to time spent picking up and completing rides, and inactive time includes waiting for ride requests (while connected to the application). At the market level, we observe average active and inactive time connected to the Uber application for all drivers eligible for the ASU program but did not enroll. We also observe average weekly earnings for this same group. For our primary analysis, we calculate average market-level earnings (often denoted as “market level active hourly pay, currently eligible”) as the ratio of average weekly earnings to average active hours for those eligible drivers who did not enroll in the program. For robustness, we also re-conduct our analysis using a measure of market-level earnings that is equal to the ratio of average earnings to average total hours (i.e., the sum of active and inactive hours). The results of the paper are both qualitatively and quantitatively consistent, regardless of the definition of market-level pay. We prefer active hourly pay because we think this is the more salient measure of compensation for drivers. There is substantial variation in market-level weekly hourly pay, where the standard deviation of this variable is \$5.5 in the sample. In addition, there is considerable variation across cities, as well as within cities across weeks. The standard deviation in weekly hourly pay across cities is \$4.3, and within cities across weeks is \$3.5.

B Market-Level Pay and Hotel Occupancy

To understand what drives the variation in market-level pay for our main results, we collected additional data on hotel vacancies for a subset of the Uber markets. The data come from STR Inc., a hotel analytics company founded in 1985 as Smith Travel Research, the recognized leader in hotel industry analytics and provides market data, including supply, demand, and market share information on a global scale. For our purposes, STR provides comprehensive daily hotel occupancy data from 2018 to 2023 across 95 Metro Markets. Our analysis's primary variables of interest are daily occupancy rates for each market. To convert the daily occupancy data to a weekly frequency, we take the average occupancy rate from Monday through Sunday of each week. However, the results are robust to alternative constructions. We can merge data for 69 of the 111 unique Uber markets, which corresponds to roughly 94% of the observations included in the results for Table 7. With these merged data, we estimate alternative specifications of the following regression:

$$MarketPay_{lt} = \beta_0 + \beta_1 Occupancy_{lt} + \beta_2 Weather_{lt} + \psi_l + \rho_m + \varepsilon_{lt}, \quad (B.1)$$

where l is the labor market, t is the week, the occupancy variable is the mean from Monday through Sunday of the given week, and ψ_l and ρ_m are city and month-year fixed effects. Results reported in Table B.1 show that the market-level weekly pay variable increases as the occupancy rate increases despite including a robust set of controls. To ease interpretation, note that a one standard deviation increase in the occupancy rate (17.5) leads to a 0.25 standard deviation increase in the market-level pay ($17.5 \times .073 = 1.28$ and $1.28/5 = .25$). Overall, these findings suggest that the variation in market pay we use for our analysis responds to shocks to demand.

Table B.1: Relationship between Market-Level Pay and Hotel Occupancy Rate

	(1)	(2)	(3)
Hotel Occupancy Rate _{it}	0.067*** (0.003)	0.074*** (0.003)	0.073*** (0.003)
Obs.	7845	7844	7609
F-stat.	440.45	526.62	97.57
Fixed-effects	No	Yes	Yes
Precip. controls	No	No	Yes
Mean dep. var.	30.07	30.07	29.9
Std. dev. dep. var.	5.11	5.11	5
Mean occ. rate	53.53	53.53	53.48
Std. dev. occ. rate	17.62	17.62	17.5

Note: The dependent variable is market-level pay. The variable “hotel occupancy rate” denotes the weekly mean occupancy rate from Monday through Sunday. The hotel occupancy rate data come from STR Inc., a hotel analytics company founded in 1985 as Smith Travel Research. “Precip. controls” denote controls for weather conditions. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

C Addressing Concerns of External Validity

C.1 Descriptive Comparison of Course-Enrollment and Non-Enrollment Weeks

While our main results should be narrowly interpreted as short-run intensive-margin responses, the following table provides a descriptive comparison of weekly Uber outcomes in weeks when drivers are and are not enrolled in ASU courses. Specifically, Table C.1 reports results from regression models where we regress a series of weekly labor market outcomes (i.e., hours worked, total pay, and hourly pay) on indicator variables for the number of ASU courses a driver is enrolled in, in a given week. In these specifications we pool all weeks of driving behavior, so the comparison value of courses is zero (i.e., weeks when drivers are not enrolled at all). We also include individual and month-year fixed effects.

These results suggest that in weeks where a student is enrolled in two courses, they work 2.87 hours less and earn roughly \$60 less than they do in weeks where they are not enrolled in any courses. For reference, the average number of courses taken during a week while enrolled in any ASU courses is 1.95. Interestingly, we find that, conditional on driving any hours, there is little economic difference in the hourly pay that drivers earn when they are and

Table C.1: Relationship between weekly Uber outcomes and academic course load

	Uber hours	Total pay	Hourly pay
1 course	-1.919*** (0.480)	-37.400*** (12.454)	-0.119 (0.216)
2 courses	-2.873*** (0.449)	-59.997*** (12.005)	0.472** (0.205)
3 courses	-3.355*** (0.801)	-90.137*** (20.426)	0.213 (0.552)
4 or more courses	-4.639*** (1.506)	-128.887*** (47.377)	-0.625 (0.936)
Obs.	169863	169863	142122
Mean dep. var.	21.45	474.66	22.19
Std. dev. dep. var.	18.28	447.43	8.86
Adjusted R ²	.42	.39	.5
Individual and month-year fixed effects	Yes	Yes	Yes

Note: This table reports estimates from regressions of weekly Uber outcomes on indicators for the number of ASU courses in which an Uber-student is enrolled during the week. The omitted category is weeks in which the driver is not enrolled in any ASU courses. “Uber hours” denotes total hours connected to the Uber platform, including both active and inactive time. “Total pay” is the sum of organic pay, incentive pay, and tips. “Hourly pay” is total pay divided by total Uber hours and is therefore observed only in weeks with positive Uber hours. The “4 or more courses” category combines weeks with four or more enrolled courses. All specifications include individual and month-year fixed effects. Standard errors, clustered at the driver level, are reported in parentheses. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

are not taking courses, possibly suggesting that the course flexibility allows them to focus their driving on periods when earning potential is high. Though not causal, these estimates provide a useful descriptive benchmark for the weekly opportunity cost of course enrollment. Reassuringly, these estimates are qualitatively consistent with the main results presented in Section 6.

C.2 Uber-Students Do Not Smooth Study Hours more than Non-Uber Peers

A potential concern with our findings is that Uber-students may be positively selected on their ability to smooth coursework over the semester or manage their time more effectively than other online students. If that were the case, the estimates we obtain in this setting might have more limited relevance for the broader population of online learners. To assess this issue, we compare the within-semester pattern of study time for Uber-students and their non-Uber peers.

For each student-course pair, we construct two measures of within-semester dispersion in weekly Canvas hours: the standard deviation and the interquartile range. We then regress each measure on an indicator for being an Uber-student, both without controls and in specifications that include course, term, and session fixed effects, so that comparisons are made within the same course offering. The results presented in Appendix Table C.2 show no evidence that Uber-students are smoothing their study time more effectively across weeks than their classmates. To the contrary, the Uber-student coefficient is positive and statistically significant for both dispersion measures in the fixed-effects specifications, indicating greater week-to-week variability in study time among Uber-students.

This analysis alleviates the concern that our findings are driven by a uniquely selected group of students who are inherently better at smoothing coursework across weeks. Instead, the evidence is more consistent with Uber-students taking greater advantage of the flexibility offered by online courses, possibly due to the variability in the opportunity cost of their time driven by within-week shifts in labor demand.

Table C.2: Within-Semester Variability in Study Time, Uber vs. Non-Uber Students

	Std. Dev. of Weekly Canvas Hours		IQR of Weekly Canvas Hours	
	(1)	(2)	(3)	(4)
Uber student	0.397*** (0.027)	0.274*** (0.012)	0.445*** (0.034)	0.308*** (0.018)
Observations	365,069	364,996	391,246	391,157
Mean dep. var.	1.00	1.00	1.16	1.16
Std. dev. dep. var.	0.85	0.85	1.03	1.03
Adjusted R^2	0.003	0.470	0.003	0.430
Fixed effects	No	Yes	No	Yes

Notes: The unit of observation is a student–course pair. In columns (1) and (2), the dependent variable is the standard deviation of weekly Canvas hours across weeks within a course. In columns (3) and (4), the dependent variable is the interquartile range (IQR) of weekly Canvas hours across weeks within a course. The indicator “Uber student” equals one for students in the ASU–Uber sample and zero for other ASU Online students. Columns (2) and (4) include course, term, and session fixed effects, so comparisons are made within the same course offering. Standard errors are reported in parentheses. Positive coefficients indicate greater within-semester variability in weekly study time among Uber students relative to non-Uber peers. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

D Decomposing the Cost of Effort

In this appendix, we attempt to decompose the marginal cost of study and work effort by quantifying what proportion of these costs can be attributed to (negative) complementarities arising from performing both activities simultaneously. For example, we aim to determine what share of the total weekly marginal cost of driving can be explained by school effort exerted in a given week since the cost of an additional hour of driving could be more daunting if an individual has been studying many hours during the week. To this end, we take advantage of the fact that it is possible to estimate reduced-form specifications closely linked to our analytical model's first-order conditions (presented in Section 4) and, therefore, compute the following ratios: $\frac{\lambda_{lc}}{\lambda_l}, \frac{\lambda_{lc}}{\lambda_c}$.⁴³ Then, we can use them jointly with the expressions for the marginal costs derived from our analytical framework to separately identify the share of the marginal cost that comes directly from exerting effort in each activity and from performing both activities simultaneously.

To do so, first consider Equation (4) that relates the optimal work effort in a given week e_{ilt}^* to a set of structural parameters and observable data moments governing the costs and benefits of studying and working.⁴⁴ In order to map Equation (4) to the data, we re-express it as:

$$e_{ilt}^* = \frac{(\alpha - 1)\lambda_{lc}\beta_1}{4\lambda_c\lambda_l - \lambda_{lc}^2} + \frac{(\alpha - 1)\lambda_{lc}\beta_5\varsigma_t}{4\lambda_c\lambda_l - \lambda_{lc}^2} + \frac{2\alpha\lambda_c w_t}{4\lambda_c\lambda_l - \lambda_{lc}^2} + \frac{(\alpha - 1)\lambda_{lc}\beta_3 A b_i}{4\lambda_c\lambda_l - \lambda_{lc}^2}. \quad (\text{D.1})$$

With Equation (D.1) in hand, we then estimate a slight variation of our primary specification in Equation (6) of the following form:

$$e_{ilt} = \gamma_0 + \gamma_1\varsigma_{it} + \gamma_2 w_{it} + \varphi_i + \vartheta_T + \varepsilon_{it}, \quad (\text{D.2})$$

where e_{ilt} denotes hours worked, ς_{it} denotes course demand across all courses in week t , w_{it} is the market wage available to individual i in week t , and φ_i and ϑ_T are individual and term fixed-effects, respectively.⁴⁵ Given this setup, we assume that each of the estimands of

⁴³See Equation (2) for definition of the λ 's.

⁴⁴The marginal benefit of studying an extra hour is the marginal improvement in grades that an hour of studying generates (weighted by $(1 - \alpha)$), and the marginal benefit of working is the market wage (weighted by α).

⁴⁵As in our previous analysis we define ς_{it} as the average study hours of course peers.

Equation (D.2) corresponds to our structural parameters as follows:

$$\gamma_1 = \frac{(\alpha - 1)\lambda_{lc}\beta_5}{4\lambda_c\lambda_l - \lambda_{lc}^2} \quad (D.3)$$

$$\gamma_2 = \frac{2\alpha\lambda_c}{4\lambda_c\lambda_l - \lambda_{lc}^2}. \quad (D.4)$$

A similar exercise for optimal study effort also yields a mapping from estimands of the estimating equation to the structural parameters. To be consistent with the analytical framework, we add study effort across all courses in which the student is enrolled in each period:

$$\sum_c^C e_{ict} = \pi_0 + \pi_1 s_{it} + \pi_2 w_{it} + \varphi_i + \vartheta_T + \epsilon_{it},$$

where

$$\pi_1 = \frac{2(1 - \alpha)\lambda_l\beta_5}{4\lambda_c\lambda_l - \lambda_{lc}^2} \quad (D.5)$$

$$\pi_2 = -\frac{\alpha\lambda_{lc}}{4\lambda_c\lambda_l - \lambda_{lc}^2}. \quad (D.6)$$

Though directly identifying any given structural parameter is not possible from Equations (D.3) - (D.6), we can use them jointly with the reduced-form estimates to recover the following ratios:⁴⁶

$$-\frac{2\gamma_1}{\pi_1} = \frac{\lambda_{lc}}{\lambda_l} \quad (D.7)$$

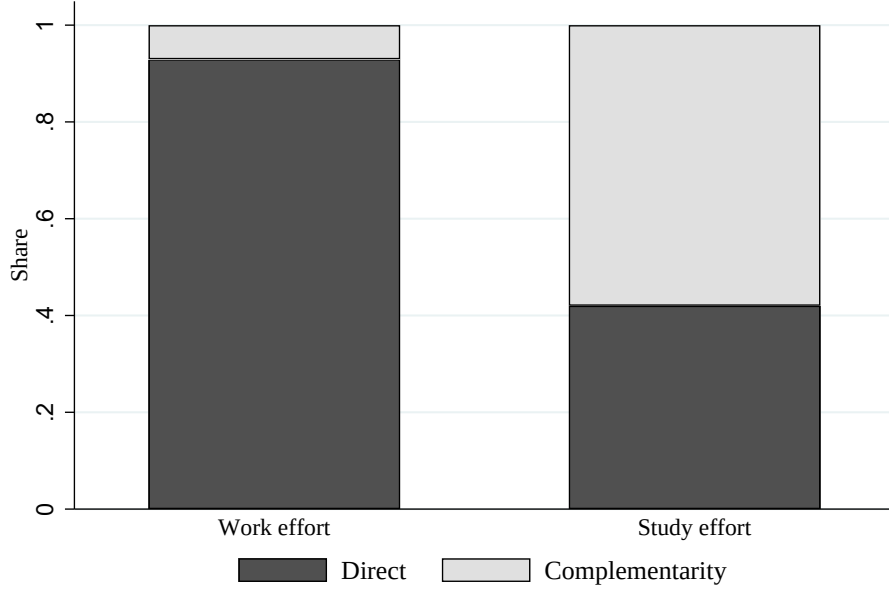
$$-\frac{2\pi_2}{\gamma_2} = \frac{\lambda_{lc}}{\lambda_c}. \quad (D.8)$$

Finally, if we combine Eqs.(D.7)- (D.8) with the following expressions of the marginal costs of working (l) and learning (c) that are derived from our analytical framework:

$$MC_l = 2\lambda_l e_{ilt} + \lambda_{lc} e_{ict} \quad (D.9)$$

⁴⁶The reduced-form coefficients (i.e., γ 's and π 's) are reported in Appendix Table G.21. To be consistent with the fact that the first order conditions only hold at interior points, in this analysis we limit our estimating sample to individual weeks with strictly positive Canvas and Uber hours. Even though we have slightly altered our specification and our estimating sample, it is worth noting that the coefficient estimates from this specification are qualitatively and quantitatively similar to those estimated from our primary specifications and presented in Tables 6 and 7.

Figure D.1: Contributions of direct effects and complementarities to marginal costs



Notes.—Each bar plot reports the share (out of 1) of the marginal cost of a given type of effort (i.e., work or study) that is directly attributable to exerting that type of effort, and that comes from the negative complementarity between exerting both types of effort simultaneously. Shares are determined at the sample average of weekly hours working and studying.

and

$$MC_c = 2\lambda_c e_{ict} + \lambda_{lc} e_{ilt}, \quad (\text{D.10})$$

then, we can uncover the proportion of MC_l and MC_c (conditional on different levels of effort) that correspond to the direct cost of exerting each activity and of exerting both activities at the same time (i.e., complementarity costs).⁴⁷

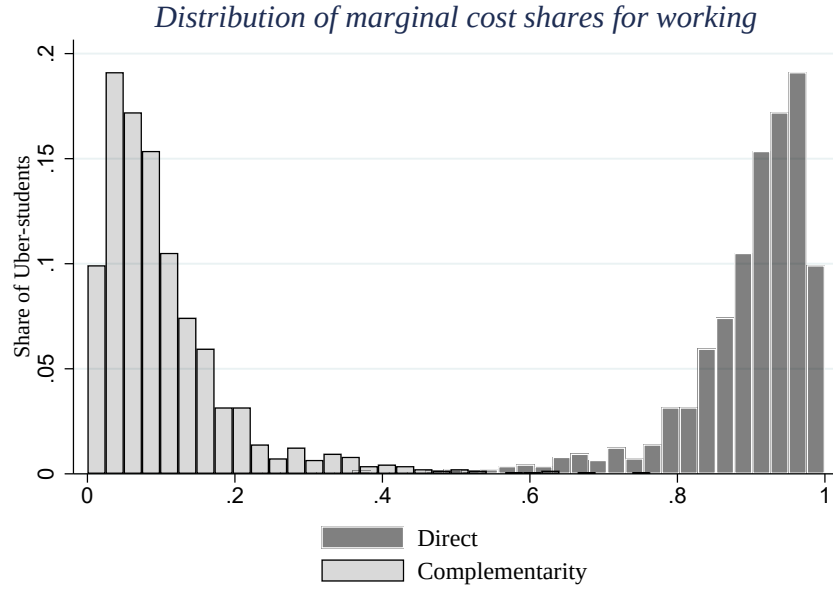
Figure D.1 presents the proportion of the marginal costs arising from complementarity effects (i.e., performing both activities simultaneously) and direct effects, where we condition on the average weekly hours of the working and learning population. The results indicate that, at the average level of work and study hours, roughly 7% of the total marginal cost of working is attributable to additional costs introduced when engaging in both activities simultaneously. Interestingly, this share is much larger for learning activities, where roughly

⁴⁷The average study and work hours for the estimating sample in this exercise are somewhat higher than the overall sample. This is, in part, because we have dropped observations with zero hours. For reference, the average study hours for this sample is 3.38 hours per week (across all courses), and the average driving hours is 22.55.

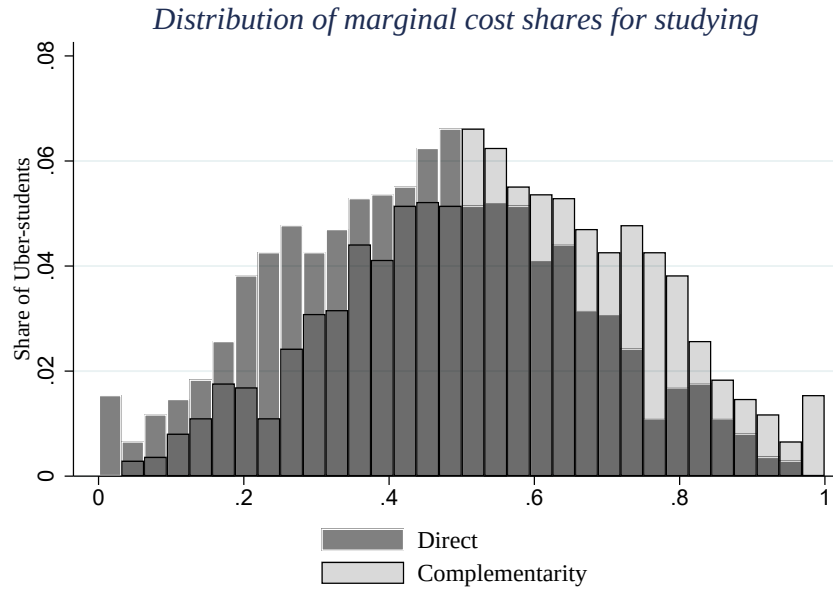
58% of the total marginal cost is due to these negative complementarities. However, this result is mainly driven by the fact that, in our sample, Uber-students spend much more time driving rather than engaging in learning activities. Finally, we find that $\frac{\lambda_l}{\lambda_c} = 0.35$, suggesting that the disutility cost of studying is approximately three times larger than from working.

In addition to analyzing the relative shares for average work and study hours, we can also uncover their distribution based on each individual's average driving and studying behavior. Figures D.2a and D.2b plot the distribution of the share of marginal costs of working and studying, respectively, that is attributable to the direct costs of each activity and due to the negative complementarities across activities. These figures reflect substantial heterogeneity in the exact source of marginal costs for these two activities (driven by the combination of hours chosen), particularly for learning activities.

Figure D.2: Distribution of Direct Marginal Costs Shares for Each Activity



(a) Working



(b) Studying

Notes:—Panel (a) plots the distribution of the share of marginal costs of working that is attributable to the direct costs of working (dark grey bar, white outline) and due to the negative complementarities between working and studying (light grey bar, black outline), and Panel (b) is analogous but for studying. A key assumption behind this exercise is that the structural parameters of our model are constant across the population but that individuals still have variation in their optimal driving behavior based on differences in earnings and course requirements. These differences yield variation in the relative contribution of direct effects and complementarities to the marginal cost of each activity.

E Interpreting the Dissimilarity Index

To help contextualize the dissimilarity index results, we first estimate the following regression equation:

$$DissimilarityIndex_{it} = \alpha_0 + \alpha_1 Class_{it} + \alpha_2 MarketPay_{it} + \alpha_3 Weather_{it} + \tau_i + \delta_m + \epsilon_{it}, \quad (E.1)$$

where $DissimilarityIndex_{it}$ is the standardized (mean zero, standard deviation one) dissimilarity index for person i in week t , $Class$ is an indicator for being enrolled in an ASU course, and τ_i and δ_m are person and month-year fixed effects. Thus, the coefficient of interest, α_1 reflects the average deviation from a driver's standard driving schedule across all course weeks. We estimate $\hat{\alpha}_1 = 0.26$.

Next, we find a random perturbation in non-course week driving behavior that can generate a similar point estimate. To do this, we randomly alter the driving hours of 30% of the observations by, first, drawing two random numbers between 1 and 28 (where each number corresponds to one of the 28-hour buckets for the week). We then add 10% of the given driver's average driving time to each of those buckets. To help illustrate, consider a driver that drives approximately 20 hours per week in non-course weeks. For this driver, we would add 2 hours to two different random hour bins (e.g., Monday 12 PM-6 PM and Thursday 6 AM-12 PM). Then, with this new variable, we recalculate the dissimilarity index and estimate the following regression limited to non-course weeks:

$$DissimilarityIndexRand_{it} = \beta_0 + \beta_1 Rand_{it} + \beta_2 MarketWage_{it} + \beta_3 Weather_{it} + \tau_i + \delta_m + \epsilon_{it}, \quad (E.2)$$

where the LHS variable is the randomized index (standardized), and $Rand$ is an indicator denoting if the observation was treated by randomization. Doing this, we estimate $\hat{\beta}_1 = 0.253$. Therefore, we interpret these results as suggesting that course demands lead to a similar deviation in typical driving behaviors as randomly altering non-course week driving schedules by 20%. It is worth noting that estimating a similar regression but with the true dissimilarity index on the LHS yields no relationship (as expected).

F Students' Willingness to Pay

We propose a straightforward model of expected utility for enrollment, as outlined in Aucejo et al. (2023), that allows us to derive measures of willingness to pay (WTP).⁴⁸ Consider the utility U_{is} that student i derives from enrolling under a specific state or scenario, denoted as s . This utility can be expressed as $U_{is} = u_i(X_s) + \epsilon_{is}$, where $u_i(X_s)$ represents the preferences of individual i for a set of characteristics X_s associated with scenario s . These characteristics might include factors like the mode of class delivery (in-person or online), the nature of work schedules (flexible or fixed), and the cost of education. The term ϵ_{is} captures an additional, scenario-specific component of the student's preference. We define ϵ_{is} as $\epsilon_{is} = \delta_i + \varepsilon_{is}$, where δ_i represents the unobserved, scenario-independent utility component for individual i , and ε_{is} corresponds to an idiosyncratic taste shock. In choice models, it is common to assume that ε_{is} is an independent and identically distributed (i.i.d.) Type I extreme value variable, independent of the preferences $u_i(X_s)$.⁴⁹ For estimation purposes, we assume that these errors are independent across individuals and conform to a Type I extreme value distribution. Therefore, the probability of student i choosing to enroll in college under scenario s can be modeled as:

$$ProbEnr_{is} = \frac{\exp(u_i(X_s) + \delta_i)}{1 + \exp(u_i(X_s) + \delta_i)}. \quad (F.1)$$

If we adopt a linear and separable utility specification, the utility function $u_i(X_s)$ can be parameterized as:

$$u_i(X_s) = \alpha_1 \cdot OnlineClass_s + \alpha_2 \cdot FlexibleWork_s + \alpha_3 \cdot Cost_s, \quad (F.2)$$

where $OnlineClass_s$ and $FlexibleWork_s$ denote indicators for online classes and flexible work schedules in scenario s , while $Cost_s$ refers to the student-specific university fees outlined in

⁴⁸For a detailed discussion on the validity of this approach and the commonly imposed identification assumptions, see Aucejo et al. (2023).

⁴⁹In our context, the independence of irrelevant alternatives (IIA) assumption is not a concern as students face only two options: enroll or drop out. ε_{is} represents resolvable uncertainty, meaning uncertainty due to unknown factors in the hypothetical survey scenario, which will become known at the time of actual decision-making.

the scenario.

We estimate this model using the fractional response approach (Papke and Wooldridge, 1996), which accommodates dependent variables within the 0 to 1 range. This method is particularly effective for handling data extremes (0 or 1 values) and is robust in the presence of censored data. The model's parameters are estimated using a quasi-maximum likelihood approach, with the Bernoulli log-likelihood function defined as:⁵⁰

$$l_{is}(\alpha, \delta) = ProbEnr_{is} \log \left(\frac{\exp(u_i(X_s) + \delta_i)}{1 + \exp(u_i(X_s) + \delta_i)} \right) + (1 - ProbEnr_{is}) \log \left(1 - \frac{\exp(u_i(X_s) + \delta_i)}{1 + \exp(u_i(X_s) + \delta_i)} \right). \quad (F.3)$$

Since the model parameters do not have a direct economic interpretation, we calculate the willingness-to-pay (WTP) for online learning as follows:

$$WTP_{OnlineClass} = -\frac{\alpha_1}{\alpha_3} \quad (F.4)$$

Similarly, the willingness to pay for flexible work schedules is determined as follows:

$$WTP_{FlexibleWork} = -\frac{\alpha_2}{\alpha_3} \quad (F.5)$$

Table F.1 shows WTP estimates for the whole sample, Uber-students, and class peers. Results indicate that students' value for work flexibility (relative to a fixed schedule) is \$4,018, while for online learning (relative to in-person classes) is \$10,243. These estimates suggest a strong willingness to pay for flexibility in learning and working formats, though the WTP for flexible working schedules is lower than for online learning. Finally, the ratio (WTP flexible work/WTP online classes) is higher for Uber-students which is expected given that they chose to have more flexible work schedules.⁵¹

⁵⁰For fractional data, the Bernoulli quasi-maximum likelihood estimator is efficient among a class of estimators that includes all quasi-maximum likelihood estimators in the linear exponential family and weighted non-linear least squares estimators.

⁵¹Survey responses indicate that Uber-students rate their work flexibility as 6.12 on average (on a scale of flexibility from 1 - not flexible at all, to 7 - extremely flexible), while their class peers rate their work flexibility as 3.85.

Table F.1: WTP Estimates for Flexible Work and Learning Schedules

		(1)		(2)		(3)	
	N	WTP Flex- ible Work	p-value	WTP On- line Classes	p-value	Work/Study WTP ratio	p-value
Uber-Students	243	2685*** (234)	0.000	5893*** (271)	0.000	0.46*** (0.04)	0.088
Class peers	344	6933*** (568)		18239*** (1176)		0.38*** (0.02)	

Note: Willingness-to-pay reported in dollars per year. Standard errors in parentheses derived via delta method. The p-value denotes whether differences in WTP between groups (i.e., Uber-students vs. class peers) are statistically significant. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

G Appendix Tables

Table G.1: Correlations of Canvas activity measures for Uber-students

	Active days	Logins	Clicks	Minutes online
Active days	1			
Logins	0.826***	1		
Clicks	0.639***	0.756***	1	
Minutes online	0.568***	0.715***	0.847***	1
Observations	71907			

Note: Table reports correlation coefficients across each metric of weekly Canvas activity. “Active days” corresponds to the number of days in the week with non-zero Canvas activity, “Logins” refer to the number of distinct times a student logged onto the course Canvas page, “Clicks” reflect the number of distinct clicks that were made on various course page links, and “Minutes online” refer to the number of minutes the student spent on the course Canvas page. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.2: Weekly Canvas activity by student group, re-weighted control group

	(1)		(2)		(3)		(4)
	Uber-Students Active in Uber		Uber-Students Inactive in Uber		Class Peers		Col (1) - (3)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Difference
Activity							
Days connected to Canvas per course	3.57	(2.303)	3.75	(2.274)	3.59	(2.241)	-0.02*
Clicks per course	75.97	(89.328)	79.04	(85.571)	69.14	(77.842)	6.83***
Hours online per course	1.57	(2.115)	1.61	(2.096)	1.39	(1.820)	0.17***
Outcomes							
Assignments submitted	1.97	(4.140)	1.99	(2.920)	2.03	(3.999)	-0.06***
Assignment share (%)	90.19	(27.587)	90.55	(26.809)	89.93	(27.767)	0.26
Point share (%)	79.40	(29.312)	80.13	(28.662)	79.49	(29.279)	-0.09
Observations	43527		12201		125649		

Note: The “Class Peers” control group is re-weighted using coarsened exact matching on observables to be more similar to the Uber-student sample. All statistics reported in the table are at the weekly level. “Uber-Students: Active in Uber” denotes weeks in which the student shows positive driving hours and is enrolled in ASU classes. “Uber Students: Inactive in Uber” denotes weeks in which the student shows zero driving hours but is enrolled in ASU classes. “Class peers” correspond to students who are enrolled in the same classes as the Uber students. Assignment and point shares refer to the total number of assignments submitted and points earned relative to the number of assignments due and points possible, respectively. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.3: Student degree progress by enrollee type

	Uber-Students		Class Peers		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	
Cumulative GPA	2.62	(1.386)	2.94	(1.091)	-0.33***
Credits to date	92.56	(54.110)	87.80	(46.490)	4.76***
Observations	1540		64248		65788

Note: Means are reported for each variable measured at the most recent term in the data for each student. “Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students enrolled in the same classes as the Uber students. “Cumulative GPA” refers to the student’s cumulative grade point average at their most recent observed term and “Credits to date” denotes the total number of academic credits on a student’s transcript including transfer credits. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.4: Course-level summary statistics, re-weighted control group

	(1)		(2)		(3)
	Uber Students		Class Peers		Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Difference
Activity					
Total hours on Canvas	14.63	(12.734)	13.00	(11.589)	1.631***
Assignments submitted	18.38	(16.892)	19.04	(17.710)	-0.663**
Outcomes					
Passed course	0.76	(0.427)	0.78	(0.413)	-0.021***
Numeric course grade	2.49	(1.664)	2.55	(1.619)	-0.062**
Observations	6591		14527		

Note: “Uber-Students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students that are enrolled in the same classes as the Uber students. The “Class Peers” group is re-weighted by coarsened exact matching on observables to be more similar to the sample of Uber-students. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.5: Correlations of Uber activity measures for enrolled Uber-students

	Total pay (100s)	Incentive pay (100s)	Tips (100s)	Completed trips	Hours online
Total pay (100s)	1				
Incentive pay (100s)	0.671***	1			
Tips (100s)	0.517***	0.296***	1		
Completed trips	0.887***	0.580***	0.534***	1	
Hours online	0.840***	0.408***	0.499***	0.833***	1
Observations			22999		

Note: Table reports correlation coefficients across each metric of weekly Uber activity. “Total pay” corresponds to the driver’s total weekly earnings from Uber, “Incentive pay” refers to pay received due to specific incentive programs (e.g., completing a target number of trips), “Tips” reflect the weekly earnings received in formal passenger tips, and “Completed trips” and “Hours online” denote the total number of trips completed and hours spent on the Uber application, respectively. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.6: Sample comparison

	Uber-Students		Class Peers	
	Survey	Admin	Survey	Admin
<i>Demographics</i>				
Age	38.78	39.25	30.64	28.51
Female	0.37	0.15	0.62	0.61
White	0.47	0.36	0.58	0.58
Black or Hispanic	0.37	0.48	0.27	0.29
<i>Academic Profile</i>				
STEM degree	0.42	0.40	0.55	0.30
Observations	243	1540	344	64248

Note: “Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students that are enrolled in the same classes as the Uber students. “Transfer status” is an indicator variable denoting whether the student transferred from another higher-education institution and “STEM degree” indicates if the student is in a science, technology, engineering, or math degree program.

Table G.7: Robustness: Determinants of work effort, leads and lags included

	(1) Work Hours	(2) Work Hours
Sum avg. study hours all c , peers	-0.668*** (0.097)	-0.551*** (0.111)
Market level active hourly pay, currently elig.	0.217*** (0.051)	0.338*** (0.071)
Observations	21572	22141
Individual Fixed Effects	✓	
Course Bundle Fixed Effects	✓	
Month-Year Fixed effects	✓	✓
Weather controls	✓	✓
Leads and lags of indep. variables	✓	✓
Mean dep. var.	17.75	17.78
Mean avg. study hours in c , peers	2.65	2.66
Market level active hourly pay, peers	33.1	33.7

Note: The variable “Sum avg. study hours all c , peers” is a proxy for coursework that denotes the sum of average Canvas hours of non-Uber students in each course c in which the Uber-student is enrolled. The variable “Market level active hourly pay, currently elig.” denotes the average hourly pay of drivers (currently eligible) in a given week. The specification includes multiple sets of fixed effects. Course bundle fixed effects denote the set of courses taken in a given term. Weather controls account for temperatures and rains/snow. A one-week lead and lag of each of the independent variables of interest are also included as controls. Column (2) does not include individual and course bundle fixed effects, given that models with lagged dependent variables can be biased when including fixed effects. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.8: Determinants of study effort (hours), week-year fixed effects included

	Study hours in c
Avg. study hours in c , peers	1.119*** (0.027)
Sum of avg. study hours in $-c$, peers	-0.007 (0.006)
Market level active hourly pay, currently elig.	-0.002 (0.004)
Observations	55,287
Individual Fixed Effects	✓
Classroom Fixed Effects	✓
Week-Year Fixed Effects	✓
Weather controls	✓
Mean dep. var.	1.58
Mean avg. study hours in c , peers	1.27
Mean sum avg. study hours in $-c$, peers	1.55
Mean market level active hourly pay, peers	33.6

Note: The dependent variable is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU–Uber partnership, but not enrolled. Specifications include individual, classroom, and week-year fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.9: Robustness: Determinants of study effort, leads and lags included

	Study Hours in c	Study Hours in c
Avg. study hours in c , peers	1.152*** (0.032)	1.152*** (0.030)
Sum avg. study hours in $-c$, peers	0.009 (0.011)	0.023* (0.012)
Market level active hourly pay, currently elig.	-0.009** (0.004)	-0.008** (0.004)
Observations	41989	41993
Individual Fixed Effects	✓	
Classroom Fixed Effects	✓	✓
Month-Year Fixed Effects	✓	✓
Leads and lags of indep. variables	✓	✓

Note: The variable “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in c . “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c in which the student was enrolled. The variable “Market level active hourly pay, currently elig.” denotes the average hourly pay of drivers (currently eligible) in a given week. The specification includes multiple sets of fixed effects. Weather controls account for temperatures and rain/snow. A one-week lead and lag of each of the independent variables of interest are also included as controls. Standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.10: Determinants of study effort (clicks and logins)

	Weekly clicks	Weekly logins	Weekly clicks	Weekly logins
Avg. study hours in c , peer	36.868*** (5.580)	2.105*** (0.596)		
Sum of avg. study hours in $-c$, peers	-0.226 (0.373)	-0.048 (0.038)		
Assignments graded in c			2.036*** (0.464)	0.220*** (0.048)
Sum of assignments graded in $-c$			-0.531*** (0.155)	-0.036** (0.015)
Market level active hourly pay, currently elig.	-0.904*** (0.196)	-0.026 (0.019)	-1.616*** (0.204)	-0.068*** (0.017)
Obs.	55293	55293	55293	55293
Individual Fixed Effects	✓	✓	✓	✓
Classroom Fixed Effects	✓	✓	✓	✓
Month-Year Fixed effects	✓	✓	✓	✓
Weather controls	✓	✓	✓	✓
Mean dep. var.	76.79	9.01	76.79	9.01
Market level active hourly pay, peers	33.57	33.57	33.57	33.57
Avg. study hours in c , peers	1.28	1.28	-	-
Sum avg. study hours in $-c$, peers	1.55	1.55	-	-
Assignments graded in c	-	-	2.21	2.21
Sum assignments graded in $-c$	-	-	2.78	2.78
Elasticity w.r.t. peer coursework	0.615	0.286	0.059	0.054
Elasticity w.r.t. peer earnings	-0.395	-0.097	-0.508	-0.253

Note: The dependent variable in each column is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . The variable “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . “Assignments graded in c ” and “Sum of assignments graded in $-c$ ” denote the number of assignments graded in course, c , and all other courses, $-c$ for a student in a given week. The results in Columns (3) and (4) are from a 2SLS specification where “Assignments graded in c ” and “Sum of assignments graded in $-c$ ” are instrumented with the average assignments graded by non-Uber peers in c and $-c$, respectively. “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Elasticities are calculated by multiplying the relevant coefficient estimate by the ratio of mean coursework or market-level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.11: Determinants of work effort (hours), lag and lead specifications

	(1) Lag specification	(2) Lead specification
<i>Market-level hourly pay</i>		
Current week	0.232*** (0.053)	0.130*** (0.048)
Lag / lead week	-0.100** (0.046)	-0.050 (0.051)
<i>Peer study hours across all courses</i>		
Current week	-0.736*** (0.092)	-0.631*** (0.075)
Lag / lead week	-0.091 (0.060)	0.018 (0.060)
Obs.	24103	24474
Individual fixed effects	Yes	Yes
Course bundle fixed effects	Yes	Yes
Month-year fixed effects	Yes	Yes
City fixed effects	Yes	Yes
Weather controls	Yes	Yes

Note: The dependent variable is weekly Uber driving hours. Standard errors clustered at the individual level are reported in parentheses. “Market-level hourly pay” denotes the average active hourly pay in a given week among drivers eligible for the ASU–Uber partnership but not enrolled. “Peer study hours across all courses” denotes the sum, across all courses, of peers’ average weekly Canvas hours. Specifications include multiple sets of fixed effects; course-bundle fixed effects correspond to the set of courses taken in a given term. Weather controls include second-order polynomials in total weekly rainfall and snowfall. The lag and lead specification columns report point estimates from the following regression models:

$$\begin{aligned}
WorkE_{i,t} = & \beta_0 + \beta_1 CourseWork_{c,t} + \beta_2 CourseWork_{c,t+s} + \beta_3 MarketHourlyPay_{l(i),t} \\
& + \beta_4 MarketHourlyPay_{l(i),t+s} + \beta_5 Weather_{l(i),t} + \psi_i + \delta_m + \phi_b + \epsilon_{i,t},
\end{aligned}$$

where $s = -1$ for the lag specification and $s = 1$ for the lead specification. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.12: Determinants of study effort (hours), lag and lead specifications

	(1) Lag specification	(2) Lead specification
<u>Market-level hourly pay</u>		
Current week	-0.002 (0.004)	-0.010*** (0.004)
Lag / lead week	-0.006* (0.004)	-0.001 (0.004)
<u>Peer Canvas hours in same course</u>		
Current week	1.121*** (0.028)	1.165*** (0.026)
Lag / lead week	0.054*** (0.015)	0.002 (0.022)
<u>Peer Canvas hours in other courses</u>		
Current week	0.003 (0.008)	-0.005 (0.008)
Lag / lead week	-0.014* (0.008)	-0.010 (0.016)
Obs.	47954	48799
Individual fixed effects	Yes	Yes
Course fixed effects	Yes	Yes
Month-year fixed effects	Yes	Yes
Weather controls	Yes	Yes
Week-length controls	Yes	Yes

Note: The dependent variable is weekly Canvas hours. Standard errors clustered at the individual level are reported in parentheses. “Market-level hourly pay” denotes the average active hourly pay in a given week among drivers eligible for the ASU–Uber partnership but not enrolled. “Peer Canvas hours in same course” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in the course. The variable “Peer Canvas hours in other courses” denotes the sum of the average hours per week that peers spent on Canvas across all other courses the student is enrolled in at the same time. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. The lag and lead specification columns report point estimates from the following regression models:

$$\begin{aligned}
CourseE_{i,c,t} = & \delta_0 + \delta_1 CourseWork_{i,t,c} + \delta_2 CourseWork_{i,t+s,c} + \delta_3 \sum_{-c} CourseWork_{i,t,-c} \\
& + \delta_4 \sum_{-c} CourseWork_{i,t+s,-c} + \delta_5 MarketHourlyPay_{l(i),t} + \delta_6 MarketHourlyPay_{l(i),t+s} \\
& + \delta_7 Weather_{l(i),t} + \delta_8 X_{c,t} + \pi_i + \alpha_m + \rho_c + \eta_{i,c,t}
\end{aligned}$$

where $s = -1$ for the lag specification and $s = 1$ for the lead specification. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.13: Heterogeneity in the determinants of work and study hours

	Full-time driver (1)	STEM (2)	Female (3)	High-need (4)	High Incoming GPA (5)	Age ≥ 35 (6)
<i>Panel A: Work hours</i>						
Market level active hourly pay, currently elig.	0.124** (0.057)	0.138** (0.063)	0.177*** (0.057)	0.134 (0.083)	0.097 (0.071)	0.143* (0.081)
Market level active hourly pay, currently elig. $\times$ var. in top row	0.096 (0.112)	0.090 (0.115)	-0.052 (0.109)	0.046 (0.102)	0.091 (0.106)	0.025 (0.100)
Sum avg. study hours all c , peers	-0.381*** (0.090)	-0.360*** (0.092)	-0.497*** (0.094)	-0.554*** (0.132)	-0.465*** (0.115)	-0.489*** (0.118)
Sum avg. study hours all c , peers $\times$ var. in top row	-0.237* (0.137)	-0.285** (0.136)	0.166 (0.157)	0.079 (0.144)	0.027 (0.134)	-0.005 (0.132)
Observations	27404	26636	26690	27388	23902	27404
<i>Panel B: Study hours</i>						
Market level active hourly pay, currently elig.	-0.010** (0.004)	-0.007 (0.005)	-0.007* (0.004)	-0.011* (0.006)	-0.007 (0.005)	-0.010* (0.005)
Market level active hourly pay, currently elig. $\times$ var. in top row	0.006 (0.006)	-0.001 (0.006)	-0.015 (0.010)	0.003 (0.007)	-0.003 (0.006)	0.004 (0.006)
Avg. study hours in c , peers	1.041*** (0.076)	1.074*** (0.042)	1.077*** (0.061)	0.978*** (0.166)	1.023*** (0.063)	1.000*** (0.048)
Avg. study hours in c , peers $\times$ var. in top row	0.207* (0.124)	0.156** (0.063)	0.315 (0.243)	0.205 (0.198)	0.107* (0.055)	0.225*** (0.047)
Sum of avg. study hours in $-c$, peers	-0.006 (0.008)	0.002 (0.008)	-0.003 (0.008)	0.004 (0.018)	-0.019** (0.008)	-0.009 (0.008)
Sum of avg. study hours in $-c$, peers $\times$ var. in top row	-0.008 (0.013)	-0.027** (0.013)	-0.045 (0.033)	-0.017 (0.022)	0.009 (0.012)	0.001 (0.011)
Observations	55293	53731	53975	55266	48263	55293

Note: “Full-time drivers” are those that average at least 25 hours of driving per week when not enrolled in ASU classes. “High-need” indicates that Uber-students are either “High” or “Very High” need based on their FAFSA records, and “High incoming GPA” indicates that the student’s incoming GPA was at least 2.72 (the sample median). For Panel A—the dependent variable in each column is hours working on the Uber platform. “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership but not enrolled. “Sum of avg. study hours across all c , peers” denotes the sum of the average hours per week that peers spent on Canvas across all courses. For Panel B—the dependent variable in each column is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . The variable “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . All specifications include multiple sets of fixed effects, second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. In Panel A, standard errors are clustered at the individual level. In Panel B, standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.14: The effect of work and study effort on completion rates

	(1) Completed Course	(2) Completed Course
Total work hours	0.001 (0.001)	
Total study hours		0.039*** (0.005)
Obs.	4259	4259
First-stage F-statistic	23.76	73.51
Course Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.		0.81
Avg. total hours on Canvas		13.50
Avg. total Uber hours		142.16

Notes: The variable “Total work hours” denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. “Total study hours” refers to the sum of individual Uber-student’s own Canvas hours throughout a given course. Each regression model includes course (e.g., ECN 101), and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the Covid-19 pandemic. The model in Column (1) is estimated via 2SLS where “Total work hours” is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where “Total Canvas hours” is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the course level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.15: The effect of work and study effort on course pass rate

	(1)	(2)
	Passed Course	Passed Course
Total work hours	-0.000 (0.001)	
Total study hours		0.032*** (0.005)
Obs.	4259	4259
First-stage F-statistic	23.76	73.51
Course Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.		0.75
Avg. total hours on Canvas		13.50
Avg. total Uber hours		142.16

Notes: The variable “Total work hours” denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. “Total study hours” refers to the sum of individual Uber-student’s own Canvas hours throughout a given course. Each regression model includes course (e.g., ECN 101), and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the Covid-19 pandemic. The model in Column (1) is estimated via 2SLS where “Total work hours” is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where “Total Canvas hours” is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the course level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.16: Expected Graduation Probabilities and Earnings in 6 Years

	Overall	Uber-Students	Class Peers	P-values of the Difference Uber-Students - Class Peers
Prob. complete studies in 6 years	0.918	0.910	0.929	0.41
Prob. complete studies in 6 years if remained at prev. college [†]	0.677	0.621	0.704	0.05
Expected returns from finishing the degree (ratio)	1.91	1.99	1.85	0.16
Observations	692	243	344	

Note: “Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students that are enrolled in the same classes as the Uber students. “Expected returns from finishing the degree” refers to the ratio of “Expected annual income in 6 years if graduate” over “Expected annual income in 6 years if dropout today”. [†] These probabilities are calculated conditional on students who transferred institutions. P-values denote whether differences between Uber students and class peers are statistically significant.

Table G.17: Higher Education Quality Comparison - Transfer Students

	All	Uber-students	Class Peers	Diff: P-values
ASU Better Educ.	0.54	0.50	0.57	0.12
ASU Inferior Educ.	0.07	0.05	0.08	0.22
ASU Similar Educ.	0.39	0.46	0.35	0.03
Observations	423	125	256	

Note: “Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to students that are enrolled in the same classes as the Uber students. The survey question and choices were *“How would you compare the quality of education at ASU to that of your previous institution? ASU offers a better/inferior/similar education than my previous institution.”*

Table G.18: ASU Credits Earned and Graduation Rates by Cohort

Academic Year Cohort	Cohort Size	Mean ASU Only Credits Earned	Graduation Rate to Date	5-Year Graduation Rate
2018–2019	147	78.79	26.5%	23.1%
2019–2020	1079	80.38	30.7%	28.5%
2020–2021	568	83.10	23.6%	23.1%
2021–2022	386	80.61	22.9%	22.9%
Cohort-Size Weighted	2180	81.02	27.2%	25.7%

Note: “Graduation rate to date” refers to the share of students observed to have graduated through the Fall 2025 semester. “5-year graduation rate” refers to the share of students who graduated within five years of initial cohort entry. “Mean ASU Only Credits Earned” excludes transfer credits. The final row reports cohort-size-weighted averages across cohorts.

Table G.19: Determinants of work effort, excluding likely dropouts

	(1) Work Hours	(2) Work Hours	(3) Work Hours	(4) Completed Trips	(5) Completed Trips	(6) Completed Trips
Sum of avg. study hours across all c, peers	-0.686*** (0.073)	-0.683*** (0.073)	-0.677*** (0.101)	-1.286*** (0.147)	-1.280*** (0.147)	-1.337*** (0.206)
Market level active hourly pay, cur- rently elig.	0.108** (0.051)	0.115** (0.051)		0.301*** (0.112)	0.326*** (0.113)	
Obs.	25972	25907	24777	25972	25907	24777
Individual Fixed Effects	✓	✓	✓	✓	✓	✓
Course Bundle Fixed Effects	✓	✓	✓	✓	✓	✓
Month-Year Fixed effects	✓	✓	✓	✓	✓	✓
Weather controls		✓			✓	
City-Week FE			✓			✓
Mean dep. var.	18.01	17.98	18.19	36.09	36.03	36.38
Market level active hourly pay, peers	33.64	33.63		33.64	33.63	
Sum avg. study hours all c, peers	2.48	2.48	2.49	2.48	2.48	2.49
Mean hourly earnings	19.1	19.1	19.23	19.1	19.1	19.23
Mean pay-per-trip	9.76	9.76	9.83	9.76	9.76	9.83

Note: Columns (1) to (3) include as dependent variable weekly work hours, while columns (4) to (6) include completed trips. The variable “Sum of avg. study hours across all c, peers” denotes the sum of the average hours across courses that peers spent on Canvas per week. The variable “Market level active hourly pay, currently elig.” denotes the average hourly pay of drivers (currently eligible) in a given week. Likely dropouts are excluded from the estimating sample. Additional details on how we identify likely dropouts are provided in Appendix A.1. Specifications include multiple sets of fixed effects. Course bundle fixed effects denote the set of courses taken in a given term. Weather controls account for temperatures and rains/snow. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.20: Determinants of study effort, excluding likely dropouts

	Study Hours in c (1)	Study Hours in c (2)
Avg. study hours in c , peers	1.206*** (0.027)	-
Sum avg. study hours in $-c$, peers	-0.011* (0.006)	-
Assignments in c	-	0.045*** (0.015)
Sum assignments in $-c$	-	-0.012** (0.005)
Market level active hourly pay, peers	-0.005 (0.003)	-0.028*** (0.004)
Observations	51385	51385
Individual Fixed Effects	✓	✓
Classroom Fixed Effects	✓	✓
Month-Year Fixed effects	✓	✓
Weather controls	✓	✓
Mean dep. var.	1.7	1.7
Avg. study hours in c , peers	1.28	-
Sum avg. study hours in $-c$, peers	1.56	-
Assignments in c	-	2.21
Sum assignments in $-c$	-	2.78
Market level active hourly pay, peers	33.58	33.57

Note: The dependent variable in each column is weekly study hours in course c . “Avg. study hours in c , peers” is a proxy for coursework that denotes the average Canvas hours of non-Uber students in course c . The variable “Sum of avg. study hours in $-c$, peers” denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c . “Assignments in c ” and “Sum of assignments in $-c$ ” denote the number of assignments graded in course, c , and all other courses, $-c$ for a student in a given week. The results in Column (2) are from a 2SLS specification where “Assignments in c ” and “Sum of assignments in $-c$ ” are instrumented with the average assignments submitted by non-Uber peers in c and $-c$, respectively. “Market level active hourly pay, currently elig.” denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Likely dropouts are excluded from the estimating sample. Additional details on how we identify likely dropouts are provided in Appendix A.1. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Standard errors are clustered at the class level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

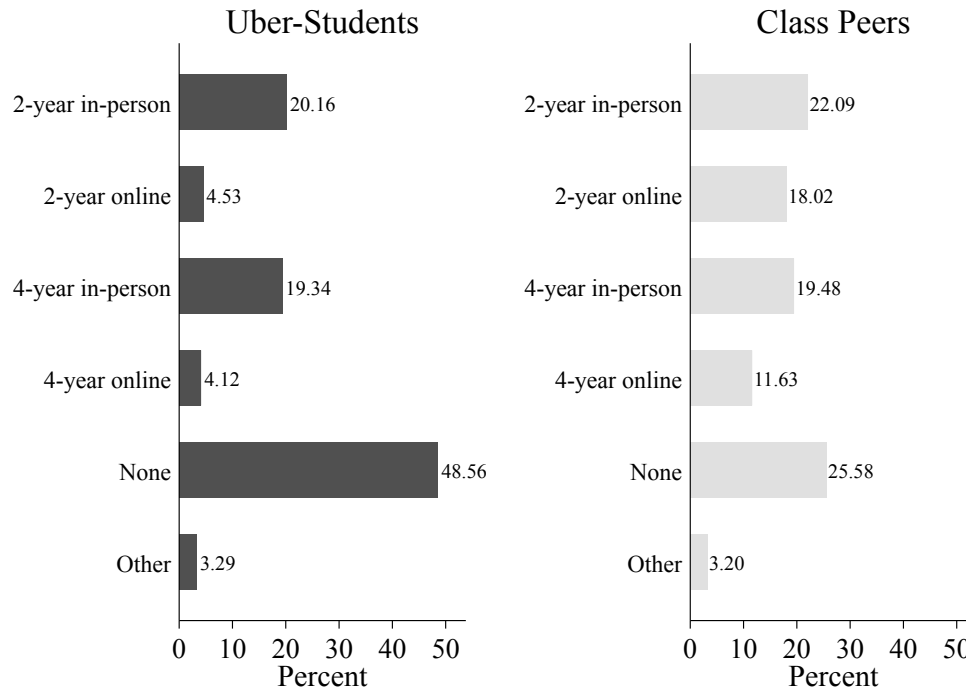
Table G.21: Intermediate coefficient estimates and model parameters

	(1)	(2)
	e_{ilt}^*	e_{ict}^*
ζ_{it}	-0.718*** (0.079)	1.311*** (0.062)
w_{it}	0.064* (0.038)	-0.012* (0.007)
Obs.	19050	19050
Implied λ_{lc}/λ_l		1.096
Implied λ_{lc}/λ_c		0.381

Note: The dependent variable Column (1) is weekly driving hours for driver i in week t and the dependent variable in Column (2) is total Canvas hours across all courses for i in t . ζ_{it} is a proxy for coursework that denotes weekly Canvas hours of non-Uber peers across all of student i 's courses. w_{it} denotes the average active hourly pay in a given week for drivers who are currently eligible for the ASU-Uber partnership but not enrolled in the same city as i . The estimation sample is restricted to person-weeks with strictly positive Canvas and Uber hours. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

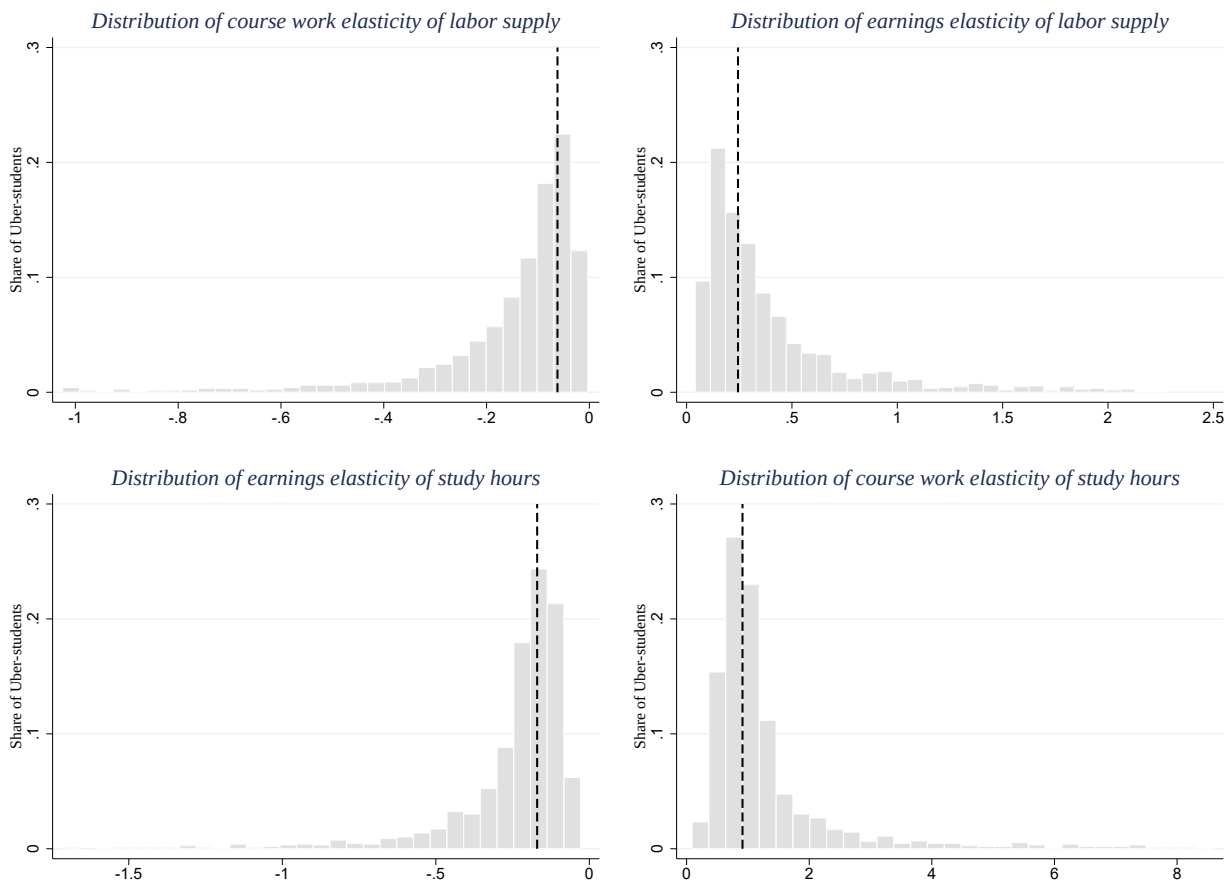
H Appendix Figures

Figure H.1: Previous Institution Attended



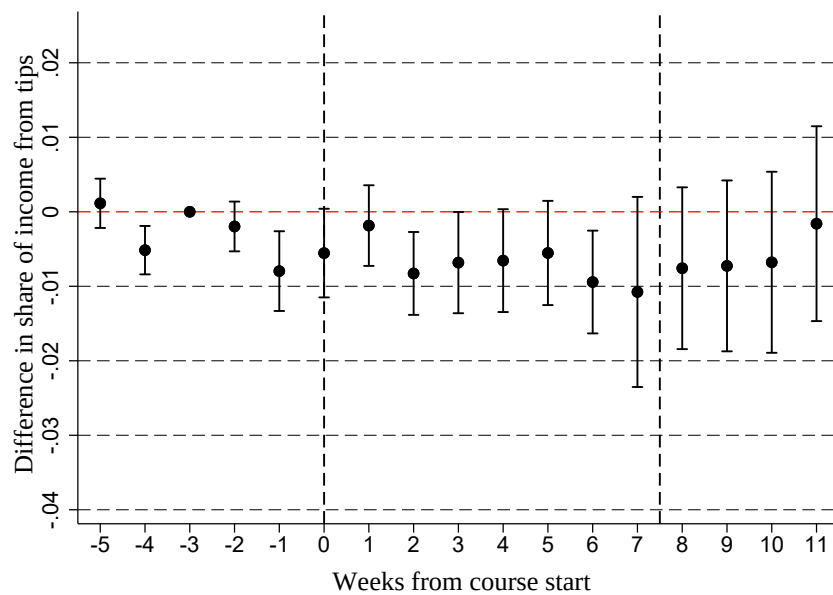
Notes:—“Uber-students” denotes ASU students that participate in the ASU-Uber partnership. “Class peers” corresponds to all other students that are enrolled in the same classes as the Uber students. There are 243 Uber-students and 344 Class peers. The corresponding survey question is: “*You indicated that you were enrolled at another institution. Which of the following best describes your previous enrollment? I attended a 4-year/2-year public/private institution and took in-person/online classes.*” We grouped the options of public and private institutions together for analysis.

Figure H.2: Distribution of Elasticities



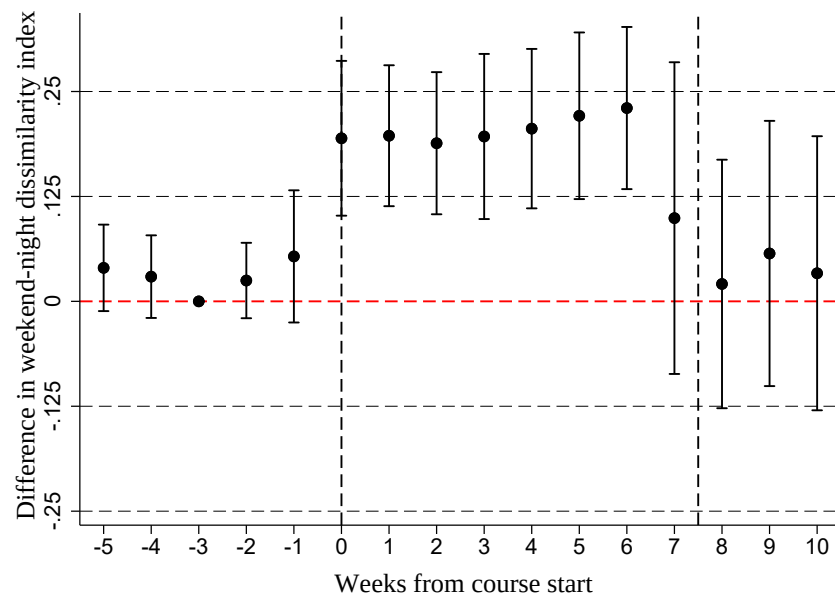
Notes:—Distribution of elasticities using individual-level means of relevant variables (i.e., average market-level pay, peer course hours, own driving, and study hours while enrolled at ASU).

Figure H.3: Variation in the share of pay coming from tips during eight-week courses



Notes:—OLS coefficient estimates from a version of Equation (1) where the dependent variable is the share of weekly earnings coming from tips and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

Figure H.4: Variation in flexibility on weekend nights during eight-week courses



Notes:—OLS coefficient estimates from a version of Equation (1) where the dependent variable is the dissimilarity index for share of driving hours allocated in Friday and Saturday afternoons and nights (i.e., 12:00PM-12:00AM) and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course.